Universidad de La Laguna
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SPECTRAL PROPERTIES OF POLARIZED SYNCHROTRON AND THERMAL DUST EMISSION FROM POWER SPECTRUM ANALYSES WITH QUIJOTE, WMAP AND PLANCK DATA

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Resumen

El fondo cósmico de microondas (CMB) constituye una de las herramientas más importantes para el estudio del Universo primitivo y de la física de la inflación. En particular, la polarización del CMB en el modo-B es sensible a las ondas gravitacionales primordiales (PGWs) generadas durante la época inflacionaria, cuya amplitud está parametrizada por el cociente tensor-escalar r. Sin embargo, la detección de este modo-B primordial no depende únicamente de la sensibilidad instrumental, sino que requiere una caracterización precisa de la emisión Galáctica polarizada que lo contamina. Las dos componentes principales que dominan la emisión polarizada del cielo en el rango de las microondas son la emisión sincrotrón, producida por electrones relativistas acelerados por el campo magnético Galáctico (GMF) y dominante a bajas frecuencias, y la emisión térmica del polvo, generada por granos de polvo interestelar alineados con el campo magnético y dominante a altas frecuencias. Ambas componentes pueden presentar fracciones de polarización de hasta el $\sim 20\%$ en ciertas regiones del cielo, superando con creces la amplitud de la señal cosmológica. El modelado y caracterización de estos fondos de radiación es por tanto un paso crítico y central en el estudio y diseño de experimentos futuros de polarización del CMB.

En este trabajo presentamos una caracterización exhaustiva de las propiedades espectrales de la emisión sincrotrón y del polvo térmico polarizados mediante un análisis del espectro angular de potencias (APS) multifrecuencia. El conjunto de datos utilizado combina los mapas DR1 del instrumento multifrecuencia de QUIJOTE (MFI) a 11 y 17 GHz, las cinco bandas de la misión WMAP (23, 33, 41, 61 y 94 GHz) observadas durante 9 años y las siete bandas de polarización del tercer lanzamiento público de Planck (PR3), que abarcan desde 28.4 hasta 353 GHz. En total, el análisis cubre el rango de frecuencias de 11 a 353 GHz, proporcionando un gran rango frecuencial clave en este análisis en polarización.

Los espectros de potencias se calculan en nueve bines de multipolo de anchura $\Delta\ell = 20$ en el rango $20 \leq \ell \leq 200$. Al combinar los autoespectros y los espectros cruzados de las 14 bandas de frecuencia, se obtienen un total de 105 espectros (14 autoespectros y 91 cruzados), lo que genera un conjunto de datos extenso y detallado para el ajuste a un modelo teórico descrito posteriormente.

El análisis se realiza sobre el cielo accesible a QUIJOTE, aplicando tres máscaras de análisis construidas a partir de la máscara de análisis de QUIJOTE combinada con cortes en latitud Galáctica de $|b| > 10^\circ$, $|b| > 15^\circ$ y $|b| > 20^\circ$. Cada máscara se apodiza con el esquema C2 de NaMaster con una escala de 5°, obteniendo fracciones de cielo efectivas de $f_{\rm sky} \approx 0.30$, 0.26 y 0.23 respectivamente. El uso de tres máscaras permite evaluar la robustez de los parámetros derivados frente a la contaminación del plano Galáctico.

El modelo de ajuste empleado consiste en una descripción simultánea de la emisión sincrotrón y del polvo térmico en el espacio armónico. La componente sincrotrón se modela mediante una ley de potencias en multipolo y en frecuencia, con amplitud A_s, índice espectral angular α_s e índice espectral en frecuencia β_s, este último gobernando la ley de escala de la distribución espectral de energía (SED) en unidades de temperatura de brillo Rayleigh-Jeans, $T_{\rm RJ} \propto \nu^{\beta_s}$. El polvo térmico se modela mediante un cuerpo negro modificado (MBB) con amplitud A_d, índice angular α_d e índice espectral β_d. Un aspecto clave del modelo es la inclusión explícita de la correlación espacial entre ambas componentes, parametrizada mediante el coeficiente $\rho \in [-1, 1]$. Los espectros cruzados codifican información complementaria sobre la geometría y coherencia del GMF a lo largo de la línea de visión. La inferencia Bayesiana de los parámetros se realiza mediante un muestreador cadena de Markov Monte Carlo (MCMC), reportando la mediana y los intervalos de credibilidad del 68% de las distribuciones a posteriori marginalizadas. Se aplican dos estrategias de ajuste complementarias: un ajuste global de ley de potencias que impone una dependencia única sobre todos los bines al multipolo, y un ajuste bin a bin independiente que permite identificar desviaciones dependientes de la escala.

Los principales resultados del ajuste de ley de potencias global son los siguientes. Para la componente sincrotrón, el índice espectral en frecuencia es $\beta_s \approx -3.0$ a -3.1 para ambos modos de polarización y las tres máscaras, en excelente acuerdo con el valor canónico de la literatura $\beta_s \sim -3.1$, extraído de análisis en el espacio real. La inclusión de los datos de QUIJOTE-MFI, que extienden la base frecuencial hasta los 11 GHz, reduce la incertidumbre en β_s en un factor de 2.6-3.0 respecto al conjunto WMAP+Planck, siendo el parámetro más beneficiado por la incorporación de QUIJOTE. El cociente B-a-E de la amplitud sincrotrón se obtiene como $r \approx 0.22$-0.23, estable para todas las máscaras y en buen acuerdo con estudios previos, que muestran valores compatibles con estos resultados. Los índices espectrales angulares α_s presentan cierta sensibilidad al rango de multipolos empleado, se muestra que el primer bin ($\ell_{\rm eff} = 29$) es especialmente relevante para determinar el modo BB. Para el polvo térmico, el índice espectral MBB es $\beta_d \approx 1.53$ en todas las configuraciones, en perfecto acuerdo con las estimaciones de Planck, mientras que los índices angulares muestran una asimetría EE-BB significativa ($> 5\sigma$). Se caracteriza además la correlación espacial cruzada entre ambas componentes, obteniendo valores pequeños pero significativamente distintos de cero ($\rho \sim 0.1$), con la correlación EE decreciendo con la latitud Galáctica y la BB estable para todas las máscaras, lo que refleja las distintas estructuras del GMF que trazan los dos modos de polarización. El análisis bin a bin corrobora todos los resultados del ajuste global sin evidencia de desviaciones dependientes de la escala angular. La robustez del análisis se verifica adicionalmente con el ajuste utilizando únicamente espectros cruzados. De esta manera se obtienen resultados plenamente consistentes con los obtenidos utilizando además los autoespectros, descartando que los resultados estén condicionados por un sesgo de ruido en los autoespectros.

Como trabajo futuro, se contemplan dos extensiones naturales de este análisis. Por un lado, se prevé incorporar la emisión anómala en microondas (AME) al modelo, aprovechando la sensibilidad única de QUIJOTE en el rango de 10-20 GHz para constatar si su contribución polarizada introduce algún sesgo en los parámetros sincrotrón recuperados. Por otro lado, se planea extender el análisis a regiones individuales del cielo de $\sim$ 5-10% de fracción de cielo cada una, con el objetivo de mapear la variabilidad espacial del índice espectral de la emisión sincrotrón β_s a nivel de espectro de potencias, proporcionando así restricciones observacionales relevantes para la separación de componentes en futuros experimentos de modo-B.

CONTENTS

1 INTRODUCTION

The Cosmic Microwave Background (CMB) provides a unique window to probe the early stages of the Universe and its structure formation. Furthermore, CMB polarization enables to probe the amplitude of primordial gravitational waves (PGWs; Ade et al., 2021) generated during the inflationary epoch (Kamionkowski, 1999; Kosowsky, 1996). These GWs give rise to the so-called B-mode of CMB polarization. In recent years, many experiments have been observing the CMB anisotropies. For instance, the Wilkinson Microwave Anisotropy Probe (WMAP; Hinshaw et al., 2013), scanning the full sky in five frequency bands between 23 and 94 GHz, and the Planck satellite (Planck Collaboration, 2020) in 9 bands, with polarization measurements in 7 of them, spanning from 30 to 353 GHz. These observations have significantly advanced our understanding of the Universe and motivated further scientific exploration, marking the beginning of a new era of high-precision cosmology with future missions centred in the study of the CMB, such as LiteBIRD (Allys et al., 2022) or other cosmology experiments.

The primordial B-mode signal might have an amplitude high enough to be within reach of current and upcoming experiments. However, its detection does not depend solely on instrumental sensitivity. A successful discovery critically relies on accurate component separation, since the cosmological signal is contaminated by polarized Galactic emission with significantly higher amplitude (Ade et al., 2021; Planck Collaboration, 2015). Component separation is therefore a central challenge in the design of future CMB experiments, directly linking the search for primordial B modes to the statistical characterization and astrophysical understanding of polarized emission from the magnetized interstellar medium (ISM).

Microwave observations also contain several astrophysical emission mechanisms other than the CMB, collectively known as foregrounds. Because the primordial B-mode signal has a very low amplitude, the removal of these foregrounds in polarization has become a crucial step for its detection. Two foreground components are known to produce strongly linearly polarized radiation: synchrotron emission, generated by relativistic cosmic-ray electrons spiraling around Galactic magnetic field (GMF) lines and dominating at low frequencies; and thermal dust emission from interstellar dust grains aligned with the magnetic field, which dominates at high frequencies. Both components can exhibit polarization fractions of up to $\sim$20% in certain regions of the sky (Planck Collaboration, 2015).

Anomalous microwave emission (AME), on the other hand, is observed to have a very low polarization fraction (Génova-Santos et al., 2017; González-González et al., 2025). If AME originates from spinning dust grains, theoretical models predict negligible polarization levels (Draine et al., 2016; Herman, D. et al., 2023). Another relevant foreground is free-free emission, produced by free electrons interacting with ionized gas; however, this mechanism generates radiation that is effectively unpolarized (Gold et al., 2011). The frequency dependence of these foregrounds in intensity and polarization at 1 degree angular scales is presented in Figure 1.

Synchrotron emission accounts for most of the radio emission from Active Galactic Nuclei (AGNs), which are thought to be powered by supermassive black holes in galaxies and quasars (Blandford & Znajek, 1977; Ishibashi, W. & Courvoisier, T. J.-L., 2011). This emission also dominates the radio continuum from star-forming galaxies, such as our own, at frequencies below $\nu \sim 30$ GHz. The relativistic electrons in nearly all synchrotron sources typically exhibit power-law energy distributions. The energy flux distribution for synchrotron radiation is given by $S_\nu \propto \nu^\alpha$, or equivalently, in terms of the brightness temperature as $T_\nu \propto \nu^\beta$, where $\beta = \alpha - 2$ (Rybicki & Lightman, 1979). This frequency dependence of the brightness temperature will be referred to as the spectral index. The spectral index depends on the energy-independent spectral energy slope s of the cosmic-ray electron spectrum, which is typically assumed to have a value of $s = -3$ in the frequency range from 10 to 30 GHz, but varying at lower frequencies (Padovani et al., 2021). Spatial variations of β reflect spatial variations of the cosmic-ray electrons and magnetic

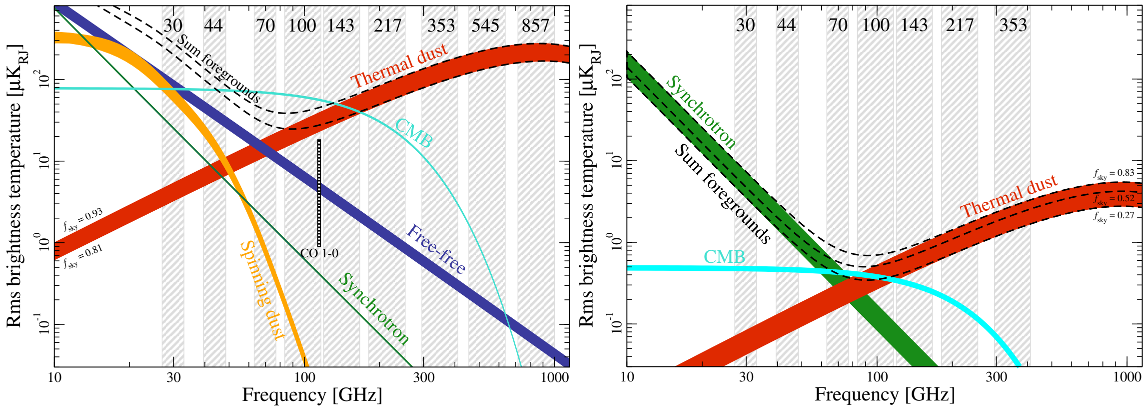

Figure 1: Frequency dependence of the main components of the submillimetre sky in temperature (left) and polarization (right; shown as $P = \sqrt{Q^2 + U^2}$). Figure taken from Aghanim et al. (2020).

field properties in the interstellar medium along the line of sight. Spatial variations of the spectral index have been mapped across large sky areas using low-frequency polarization surveys. In particular, S-PASS observations at 2.3 GHz have characterized synchrotron emission in the southern sky (Krachmalnicoff et al., 2018), while QUIJOTE-MFI measurements at 10-20 GHz have probed the northern sky (Rubiño-Martín et al., 2023). Both studies report statistically significant spatial variations of the spectral index.

At higher frequencies, the dominant polarized foreground is thermal emission from interstellar dust grains aligned with the GMF. This emission dominates the microwave sky above approximately 100 GHz and is partially linearly polarized. The frequency spectrum of thermal dust emission is well described by a modified black body (grey body), $I_d(\nu) \propto \nu^{\beta_d} B_\nu(T_d)$, where $B_\nu(T_d)$ is the black-body spectrum at dust temperature T_d. Using combined Planck and WMAP data, Planck Collaboration et al. (2015) determined the dust spectral index in both intensity and polarization, finding $\beta_d \simeq 1.59$ for $T_d \simeq 19.6\,\mathrm{K}$, while Planck Collaboration et al. (2020b) derived a mean polarization spectral index of $\beta_d = 1.53 \pm 0.02$ from PR3 maps at 353 GHz, with the dust EE and BB power spectra following power laws in multipole and exhibiting a significant E/B asymmetry. Since both dust and synchrotron polarization are aligned by the same GMF, a spatial cross-correlation between the two components is expected and has indeed been detected at the level of $\rho \approx 0.1$-0.2 (Choi & Page, 2015; Planck Collaboration et al., 2020b).

The sensitivities of WMAP and Planck alone do not allow a sufficiently accurate characterization of the polarized synchrotron signal at intermediate and high Galactic latitudes, leaving important uncertainties in the modelling of this foreground component. This is significantly improved by the QUIJOTE experiment (Rubiño-Martín et al., 2023), which observes the polarized Northern sky in the 10-40 GHz range, precisely where synchrotron emission is strongest and where Faraday rotation effects are not expected to be significant (Fuskeland et al., 2021a). Combining QUIJOTE with WMAP and Planck data, de la Hoz et al. (2023) performed one of the first reconstructions of the polarized synchrotron spectral index in real space, finding statistically significant spatial variations across the sky, with an average value of $\beta = -3.08$ and a dispersion of 0.13. More recently, Casas et al. (2025) and Adak et al. (2025) provided new constraints on the synchrotron spectral index. In particular, Adak et al. (2025) applied a semi-blind component separation method to QUIJOTE-MFI data, reinforcing the evidence for spatial variability and highlighting the importance of combining QUIJOTE observations with complementary data from WMAP and Planck in order to reduce noise contamination and improve the robustness of spectral-index estimates.

In this work, we characterize the spectral properties of polarized synchrotron and thermal dust emission through power spectrum analyses of the combined QUIJOTE MFI DR1 (11, 13, 17, and 19 GHz), WMAP 9-year (K, Ka, Q, V, and W bands), and Planck PR3 (LFI and HFI, 28.4–353

GHz) datasets. Both foreground components are modelled simultaneously by fitting a parametric spectral energy distribution (SED) model to the EE and BB angular power spectra (APS) and their cross-spectra across this wide frequency range. A key aspect of this analysis is the explicit treatment of the spatial cross-correlation between polarized synchrotron and thermal dust emission. Their cross-spectra encode complementary information about the geometry and coherence of the GMF along the line of sight (Planck Collaboration et al., 2015; Choi & Page, 2015). The analysis is carried out over the full Northern sky footprint of QUIJOTE, using three Galactic latitude masks to assess the sensitivity of the results to Galactic contamination. The resulting foreground characterization is intended to serve as legacy input for component separation in upcoming CMB polarization experiments such as LiteBIRD (Allys et al., 2022).

2 OBJECTIVES

The main objective of this work is to characterize the spectral properties of the two dominant polarized Galactic foregrounds: synchrotron and thermal dust emission, using harmonic-space (power spectrum) tools. The analysis is performed on the combined dataset of QUIJOTE MFI DR1 maps at 11 and 17 GHz (Rubiño-Martín et al., 2023), all five WMAP 9-year frequency bands (K, Ka, Q, V, and W, Bennett et al., 2013), and the three LFI (Planck Collaboration, 2016a) and four HFI polarization bands (Planck Collaboration et al., 2020a) from Planck PR3, spanning a total frequency range from 11 to 353 GHz. This broad frequency coverage enables us to simultaneously fit a parametric SED model for both foreground components.

A second key objective is the characterization of the spatial cross-correlation between polarized synchrotron and thermal dust emission. Since both foregrounds are polarized by the same GMF, their EE and BB angular power spectra are expected to be partially correlated. Measuring this cross-correlation, alongside the auto-spectra of each component, provides complementary constraints on the geometry and coherence of the GMF and improves the accuracy of foreground separation in the CMB polarization context (Choi & Page, 2015).

Our results will be compared with previous power-spectrum analyses that did not include observations below 23 GHz. Martire et al. (2022) characterized the polarized synchrotron emission using WMAP and Planck data alone, finding a spectral index of $\beta_s = -3.00 \pm 0.10$ and a B-to-E mode amplitude ratio of 0.24 ± 0.04 with masks covering a sky fraction of 0.5. Krachmalnicoff et al. (2018) obtained a steeper mean value of $\beta_s = -3.22 \pm 0.08$ when including the lower-frequency S-PASS data at 2.3 GHz in the southern sky.

For the thermal dust component, our results will be compared with those of Planck Collaboration et al. (2020b), who used Planck PR3 maps at 353 GHz to characterize dust polarization power spectra over sky regions covering 24 to 71% of the sky. They found power-law multipole indices α (parametrizing the ℓ-dependence of the power spectrum as $C_\ell \propto \ell^\alpha$) of $\alpha^{EE} = -2.42 \pm 0.02$ and $\alpha^{BB} = -2.54 \pm 0.02$, a significant E/B power asymmetry, and variations of the spectral exponents across sky regions. The dust polarization SED was well described by a modified black body with spectral index $\beta_d = 1.53 \pm 0.02$, with no evidence for frequency decorrelation between 100 and 353 GHz. For the synchrotron-dust cross-correlation, the benchmark is provided by Choi & Page (2015), who showed that a joint foreground model including the synchrotron-dust cross-correlation fits all WMAP and Planck frequency cross-spectra. These results constitute the main observational benchmarks against which our joint synchrotron-plus-dust analysis will be validated.

Beyond the immediate scientific goals, this work aims to provide robust foreground constraints and data products, including spectral index estimates and validated foreground models, that will serve as essential inputs for component separation in forthcoming CMB polarization experiments.

3 DATA

All data used in this work consist of three independent maps for each frequency band: one intensity map, corresponding to the Stokes parameter I, and two maps describing linear polarization, given by the Stokes parameters Q and U. In this work, we consider polarization data from three experiments: QUIJOTE-MFI, WMAP, and Planck, which are described below.

3.1. Q-U-I JOint TEnerife (QUIJOTE)

The QUIJOTE experiment (Rubiño-Martín et al., 2023) observes the visible sky from the Teide Observatory (latitude $+28.3°$), covering all regions with elevations greater than 30° in both intensity and polarization. The experiment consists of two telescopes equipped with three polarimetric instruments: the Multifrequency Instrument (MFI), operating in the 10–20 GHz range; the Thirty-GHz Instrument (TGI); and the Forty-GHz Instrument (FGI).

In this work we use data from the 11 and 17 GHz bands of the MFI, which have angular resolutions of 55 and 38 arcmin, respectively. The data are obtained from the Legacy Archive for Microwave Background Data Analysis (LAMBDA)[1]. The 11 GHz band is particularly useful for characterizing the polarization of low-frequency foregrounds, primarily synchrotron emission (Adak et al., 2025; Casas et al., 2025) and AME (Fernández-Torreiro et al., 2023; González-González et al., 2025).

The 13 and 19 GHz bands of the MFI are not included in this analysis, as they share the same horn pairs as the 11 and 17 GHz bands respectively, and are therefore significantly correlated (Rubiño-Martín et al., 2023). Including correlated frequency maps in the power spectrum estimation would bias the foreground parameter constraints. The two independent bands at 11 and 17 GHz are therefore sufficient to represent the QUIJOTE-MFI dataset in this work.

3.2. Wilkinson Microwave Anisotropy Probe (WMAP)

WMAP was a space-based experiment that observed the total intensity and polarization of the full sky across five frequency bands, spanning the frequency range from 23 to 94 GHz. In this work, we make use of all five frequency bands from the 9-year data release (Bennett et al., 2013): the K-band (23 GHz, $FWHM = 52.8$ arcmin), Ka-band (33 GHz, $FWHM = 39.6$ arcmin), Q-band (41 GHz, $FWHM = 31.2$ arcmin), V-band (61 GHz, $FWHM = 21.0$ arcmin), and W-band (94 GHz, $FWHM = 13.2$ arcmin). All maps are downloaded from the LAMBDA database[2].

3.3. Planck

Planck was a space-based experiment operating two complementary instruments: the Low Frequency Instrument (LFI) and the High Frequency Instrument (HFI), together observing the full sky in intensity and polarization across nine frequency channels, from 30 to 857 GHz. In this work, we use data from the third public release (PR3) of the Planck mission, available through the Planck Legacy Archive[3] (PLA).

3.3.1 Low Frequency Instrument (LFI)

The LFI instrument covered three frequency bands at 28.4, 44.1, and 70.4 GHz. We use the PR3 frequency maps from all three LFI channels (Planck Collaboration, 2016a). The original angular resolutions correspond to effective beams of 32.65, 27.94, and 13.01 arcmin at 28.4, 44.1, and 70.4 GHz, respectively. All maps are downgraded to a HEALPix[4] resolution of $N_{\rm side} = 512$ to maintain a common pixelization with the QUIJOTE and WMAP datasets.

[1] https://lambda.gsfc.nasa.gov/product/quijote/index.html
[2] https://lambda.gsfc.nasa.gov/product/wmap/current/index.html
[3] https://pla.esac.esa.int/
[4] https://healpy.readthedocs.io/en/latest/

3.3.2 High Frequency Instrument (HFI)

The HFI instrument provides six frequency bands extending up to 857 GHz. In this analysis we use the PR3 frequency maps from the HFI channels up to 353 GHz (100, 143, 217, and 353 GHz) (Planck Collaboration et al., 2020a). These four channels provide polarization information, including Stokes Q and U maps, and are therefore used in our polarization analysis. The higher-frequency HFI channels at 545 and 857 GHz are not included, as they do not provide polarization measurements. To ensure a consistent pixelization across the QUIJOTE, WMAP, and Planck datasets, all selected maps are downgraded to $N_{\rm side} = 512$.

4 METHODOLOGY

In this section we describe the methodology followed to characterize Galactic foreground emission using multi-frequency angular power spectrum analysis. The approach relies entirely on harmonic-space statistics, avoiding the complications of real-space fitting, and exploits the broad frequency coverage provided by the combination of QUIJOTE, WMAP, and Planck datasets. We first introduce the mathematical framework of angular power spectrum analysis (Section 4.1.), then describe the construction of the analysis masks (Section 4.2.), the data processing steps (beam in Section 4.3.1, unit conversion in Section 4.3.2, noise in Section 4.3.3, and colour corrections in Section 4.3.4), the simulations used to estimate uncertainties (Section 4.4.), the foreground models (Section 4.5.), and finally, the Bayesian parameter inference (Section 4.6.).

4.1. Angular power spectrum (APS) analysis

The APS is a statistical tool that describes the distribution of signal power as a function of angular scale on the sky. It is a particularly powerful instrument for studying the statistical properties of diffuse emission, since it allows one to separate contributions from different physical processes through their characteristic frequency and multipole dependencies. Rather than working directly in pixel space, all analyses presented in this work are carried out in harmonic space, where the APS constitutes the fundamental observable.

The APS is expressed in terms of the spherical harmonic coefficients $a_{\ell m}$, which encode the sky signal at each multipole ℓ and azimuthal order m. The multipole ℓ is inversely proportional to the angular scale $\theta \sim 180°/\ell$, so that large angular scales correspond to low-ℓ values, and small angular scales to high-ℓ values. For reference, $\ell = 200$ corresponds to approximately 1°, which is comparable to the angular resolution of the QUIJOTE-MFI instrument. The cross-power spectrum between two frequency maps X and Y is defined as:

$$C_\ell^{X\times Y} = \left\langle \frac{\sum_m a_{\ell m}^X (a_{\ell m}^Y)^*}{2\ell+1} \right\rangle, \tag{1}$$

where X and Y denote the two frequency bands. When $X = Y$ this reduces to the auto-power spectrum. Cross-spectra probe the signal component common to two independent observations, suppressing the contribution of uncorrelated noise. By combining auto- and cross-spectra from 14 frequency bands (2 QUIJOTE, 5 WMAP and 7 Planck channels), we form a total of 105 spectra (14 auto and 91 cross), providing broad multi-frequency coverage for constraining the foreground model parameters.

All power spectra in this work are computed using the NaMaster Python library[5] (Alonso et al., 2019), which implements the pseudo-C_ℓ method with an analytical treatment of the mode-coupling matrix induced by the sky mask. In particular, the E- and B-mode polarization fields are treated with B-mode purification (Smith, 2006), which suppresses E-to-B leakage arising from the application of a partial sky mask. The power spectra are computed in nine multipole bins of width $\Delta\ell = 20$ over the range $20 \leq \ell \leq 200$, with bin edges defined as

$$\ell \in [\ell_1^{(k)}, \ell_2^{(k)}], \qquad \ell_1^{(k)} = 20 + 20(k-1), \qquad \ell_2^{(k)} = 39 + 20(k-1), \qquad k = 1, \ldots, 9. \tag{2}$$

[5] https://namaster.readthedocs.io/en/latest/

The lower multipole limit of $\ell = 20$ is imposed by the FDEC filtering applied to the QUIJOTE-MFI maps to remove residual radio frequency interference (RFI) (Rubiño-Martín et al., 2023). This filtering suppresses large-angular-scale modes, making multipoles below $\ell \approx 20$ unreliable for a power spectrum analysis.

The effective multipole $\ell_{\rm eff}$ reported for each bin is computed by NaMaster as the weighted average of the individual multipoles within the bin:

$$\ell_{\rm eff}^{(k)} = \frac{\sum_{\ell \in \text{bin k}} w_\ell \, \ell}{\sum_{\ell \in \text{bin k}} w_\ell}, \tag{3}$$

where w_ℓ are the per-multipole weights assigned to the bandpower. In this work uniform weights are used, so $\ell_{\rm eff}^{(k)}$ reduces to the arithmetic mean of all integer multipoles in the k-th bin, yielding $\ell_{\rm eff} = 29, 49, 69, \ldots, 189$ for the nine bins.

4.2. Masks

The analysis is restricted to the sky region accessible to the QUIJOTE-MFI telescope (Section 3.1.), which operates from the Teide Observatory and therefore covers a declination band of approximately $-30^\circ < \delta < +90^\circ$. To mitigate contamination from the bright Galactic plane, three analysis masks are constructed by combining the QUIJOTE low-declination survey footprint (Rubiño-Martín et al., 2023) with Galactic latitude cuts at $|b| > 10^\circ$, $|b| > 15^\circ$, and $|b| > 20^\circ$. Each binary mask is apodized using the C2 scheme of NaMaster with an apodization scale of 5°, which tapers the mask smoothly to zero at its edges and suppresses ringing effects in harmonic space. The three masks are shown in Figure 2. The effective sky fractions after apodization are approximately $f_{\rm sky} \approx 0.30$, 0.26, and 0.23 for the 10, 15, and 20 degree cuts, respectively. The masks are defined on a HEALPix grid at $N_{\rm side} = 512$ and are identical for all frequency bands.

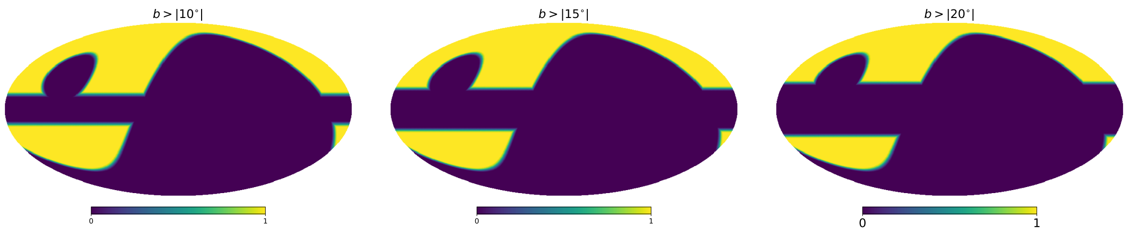

Figure 2: Analysis masks combining the QUIJOTE low-declination survey footprint with Galactic latitude cuts of $|b| > 10^\circ$ (left), $|b| > 15^\circ$ (middle), and $|b| > 20^\circ$ (right), after C2 apodization with a 5° border.

4.3. Data corrections

The raw power spectra computed from the observed maps contain systematic contributions from instrumental effects that must be accounted for before a physical interpretation can be drawn. In the ideal case, the measured C_ℓ coefficients would be proportional to the square of the theoretical spherical harmonic coefficients, $|a_{\ell m}^{\rm theo}|^2$. In practice, several multiplicative and additive corrections are applied, as described below.

4.3.1 Beam and pixel window functions

The finite angular resolution of each instrument and the pixelization of the HEALPix maps (Górski et al., 2005) introduce multiplicative window functions that suppress power at high multipoles. The pixel window function, w_ℓ, arises from the averaging of the sky signal over each pixel and depends solely on $N_{\rm side}$. For $N_{\rm side} = 512$ this correction is small but non-negligible at $\ell > 500$. The instrumental beam is characterized by the window function B_ℓ, which captures the effective angular resolution of the experiment and causes an exponential suppression of power above the beam scale.

For each frequency band, the beam window function B_ℓ is loaded from the official data products of each experiment and interpolated to the effective multipoles of the binning scheme. For QUIJOTE-MFI, the beam is provided as a tabulated array per frequency channel in a FITS file. For WMAP, it is provided as a normalized ASCII table. For Planck LFI and HFI bands, the beam window functions are taken from the respective official FITS products, with separate T, E, and B transfer functions for HFI channels.

Both cases, auto- and cross-spectra, are handled by a single correction equation. Defining the effective transfer function for band i as $\mathcal{W}_\ell^i \equiv w_\ell\, B_\ell^i$, where w_ℓ is the HEALPix pixel window function (common to all bands at fixed $N_{\rm side}$) and B_ℓ^i is the instrumental beam window function of band i, the corrected spectrum between bands i and j is:

$$C_\ell^{i\times j,\,{\rm corr}} = \frac{C_\ell^{i\times j,\,{\rm sky}} - \delta_{ij}\, N_\ell^i}{\mathcal{W}_\ell^i\, \mathcal{W}_\ell^j}, \tag{4}$$

where δ_{ij} is the Kronecker delta, N_ℓ^i is the noise power spectrum of band i (see Section 4.3.3), and the denominator deconvolves the combined effect of the pixel window and the beam of each band. The Kronecker delta ensures that the noise bias is subtracted only for auto-spectra ($i = j$); for cross-spectra between different frequency bands the noise contributions of the two detectors are statistically independent and thus average to zero in the cross-correlation, so no noise subtraction is required.

4.3.2 Unit conversion

All data are converted to a common unit of Rayleigh-Jeans brightness temperature ($\rm K_{RJ}$). The unit conversion factor between thermodynamic CMB temperature fluctuations ($\rm K_{CMB}$) and Rayleigh-Jeans temperature is

$$u(\nu) = \frac{x^2\, e^x}{(e^x-1)^2}, \qquad x \equiv \frac{h\nu}{k_B T_{\rm CMB}}, \tag{5}$$

where h is the Planck constant, k_B the Boltzmann constant, and $T_{\rm CMB} = 2.72548\,{\rm K}$ (Fixsen, 2009). This factor follows from the derivative of the Planck spectrum with respect to temperature evaluated at $T_{\rm CMB}$, $u(\nu) = \left(c^2/2k_B\nu^2\right)\left(\partial B_\nu/\partial T\right)_{T_{\rm CMB}}$. For the Planck HFI bands (100–857 GHz), this analytical expression is replaced by the official bandpass-averaged unit conversion coefficients from the Planck Legacy Archive (Planck Collaboration et al., 2020a), which account for the measured instrumental bandpass shape and the associated frequency shift. For all remaining bands (QUIJOTE, WMAP, and Planck LFI) the conversion is evaluated analytically at the nominal band-centre frequency. The unit conversion applied to the power spectrum of band pair (i, j) is $u(\nu_i)u(\nu_j)$.

4.3.3 Noise power spectrum

The instrumental noise contributes an additive bias to the auto-power spectrum that must be estimated and removed (Equation 4). The noise estimation strategy is specific to each experiment.

For QUIJOTE-MFI, the HMDM is constructed from the two half-survey maps provided in the DR1 data release (Rubiño-Martín et al., 2023), together with their associated per-pixel weight maps w_1 and w_2 (where $w_i = 1/\sigma_i^2$):

$$n = \frac{h_1 - h_2}{\sqrt{(w_1+w_2)(w_1^{-1}+w_2^{-1})}}. \tag{6}$$

This construction ensures that the noise power spectrum of the difference map matches that of the combined weighted-sum map, while containing no sky signal (Planck Collaboration, 2016a).

For Planck, the same methodology is followed but the nature of the data split differs between the two focal plane arrays. For the LFI (30-70 GHz), ring-half maps are used as h_1 and h_2, with pixel covariance maps providing the weights (Planck Collaboration, 2016a). For the HFI (100-353 GHz), the official half-mission maps from the Planck 2018 data release (Planck Collaboration et al., 2020a) are used. In both cases, the covariance-weighted HMDM is constructed using Equation 6. Since Planck maps are provided at a higher resolution ($N_{\rm side}$), the resulting HMDM is downgraded to $N_{\rm side} = 512$ after construction to match the resolution of the other datasets.

For WMAP, the official band-average maps are noise-weighted co-additions of all differential assemblies (DAs) at each frequency. No half-mission split is provided directly for the band-average maps; instead, we construct the HMDM from the temporal split available in the public data release. We use the first four years of observations (h_1, years 1 to 4) and the remaining five years (h_2, years 5 to 9). Because the two halves average over different numbers of years, the naive difference $h_1 - h_2$ does not reproduce the noise level of the full 9-year combination. We apply the analytical normalization factor that accounts for the unequal time splits:

$$n_{\rm WMAP} = (h_1 - h_2)\sqrt{\frac{t_1\, t_2}{(t_1 + t_2)^2}}, \tag{7}$$

where $t_1 = 4$ and $t_2 = 5$ are the number of years in each half, giving a normalization factor of $\sqrt{20/81}$. This ensures that the noise power spectrum of the HMDM matches that of the full 9-year map, and it is used directly as N_ℓ^i in Equation 4.

4.3.4 Colour corrections

Colour corrections account for the mismatch between the SED of the foreground emission and the spectral response of each instrumental bandpass. When a source with spectral index β is observed through a band with transmission $\tau(\nu)$, the measured flux density differs from the value at the nominal band-centre frequency by a factor that depends on β. This factor is the colour correction $\mathrm{CC}(\beta)$.

The colour correction coefficients for each band and each foreground component (synchrotron and thermal dust) are pre-computed using the `fastcc` code[6] (Peel et al., 2022), which evaluates the bandpass-weighted ratio between the assumed foreground SED and a flat (Rayleigh-Jeans) reference spectrum.

The treatment differs between foreground components because they follow different SEDs. For synchrotron emission (power law, $I_\nu \propto \nu^\alpha$), the colour correction depends only on a single spectral index α, so it can be parameterized as a 1D polynomial. The same applies to dust emission in non-HFI bands, where the modified blackbody curvature is negligible and a power-law approximation is sufficient. For these components, the coefficients are obtained using `fastcc`.

For thermal dust in the Planck HFI bands (100-353 GHz) the SED follows a modified blackbody (MBB),

$$I_\nu \propto \nu^{\beta_d}\, B_\nu(\nu, T_d), \tag{8}$$

where the Planck function $B_\nu(\nu, T_d)$ introduces a non-trivial dependence on the dust temperature T_d that cannot be absorbed into a power-law index. As a result, the HFI dust colour correction depends jointly on both β_d and T_d, and a simple 1D polynomial in α is insufficient. It is therefore computed using the `interpcc` interpolator (provided with `fastcc`), which interpolates a two-dimensional grid of pre-computed colour corrections as a function of both β_d and T_d (Peel et al., 2022). The grid is evaluated at the fixed dust temperature $T_d = 19.6$ K adopted throughout this work, reducing the 2D surface to a 1D function of β_d alone, which is then fitted with a second-order polynomial.

[6] https://github.com/mpeel/fastcc

In all cases the resulting colour correction for component $c \in \{s, d\}$ at frequency band ν is expressed as a second-order polynomial in the relevant spectral index:

$$\mathrm{CC}_c(\nu;\, \alpha) = a_0^c(\nu) + a_1^c(\nu)\,\alpha + a_2^c(\nu)\,\alpha^2, \tag{9}$$

where $\alpha = \beta_s + 2$ for synchrotron ($c = s$) and $\alpha = \beta_d + 2$ for dust ($c = d$) in Rayleigh-Jeans units. The polynomial coefficients $\{a_0^c(\nu),\, a_1^c(\nu),\, a_2^c(\nu)\}$ are specific to each component c and frequency band ν, and are derived using `fastcc` following the methodology described above.

Rather than being applied as a fixed correction to the data, the colour corrections are folded directly into the foreground model (see Section 4.5.) at each MCMC iteration, using the spectral index proposed by the sampler at that step. Each foreground component carries its own correction, applied at each frequency of the pair. The colour-corrected model terms are:

$$C_\ell^{\mathrm{sync,\,cc}}(\nu_i, \nu_j) = \frac{A_s\,(\ell/\ell_{\mathrm{ref}})^{\alpha_s}\, S_s(\nu_i)\, S_s(\nu_j)}{\mathrm{CC}_s(\nu_i;\, \beta_s)\,\mathrm{CC}_s(\nu_j;\, \beta_s)}, \tag{10}$$

$$C_\ell^{\mathrm{dust,\,cc}}(\nu_i, \nu_j) = \frac{A_d\,(\ell/\ell_{\mathrm{ref}})^{\alpha_d}\, S_d(\nu_i)\, S_d(\nu_j)}{\mathrm{CC}_d(\nu_i;\, \beta_d)\,\mathrm{CC}_d(\nu_j;\, \beta_d)}, \tag{11}$$

and for the synchrotron-dust cross-correlation, each frequency leg receives the correction appropriate for its own component:

$$C_\ell^{\mathrm{corr,\,cc}}(\nu_i, \nu_j) = \rho\sqrt{A_s A_d}\left(\frac{\ell}{\ell_{\mathrm{ref}}}\right)^{(\alpha_s+\alpha_d)/2}\left[\frac{S_s(\nu_i)}{\mathrm{CC}_s(\nu_i;\, \beta_s)}\frac{S_d(\nu_j)}{\mathrm{CC}_d(\nu_j;\, \beta_d)} + \frac{S_s(\nu_j)}{\mathrm{CC}_s(\nu_j;\, \beta_s)}\frac{S_d(\nu_i)}{\mathrm{CC}_d(\nu_i;\, \beta_d)}\right]. \tag{12}$$

Since the corrections are re-evaluated at every likelihood call, they track the spectral index throughout the chain rather than being fixed at a single approximate value. This propagates the uncertainty in the spectral index self-consistently into the corrected model, improving upon approaches where the colour correction is applied as a fixed factor computed at a mean spectral index.

4.4. Simulations

The statistical uncertainties on the measured power spectra are estimated from Monte Carlo simulations. In the corrected spectrum (Equation 4), two independent terms contribute to the total error: the uncertainty on the sky power spectrum $C_\ell^{i\times j,\,\mathrm{sky}}$, characterized by σ_{C_ℓ}, and the uncertainty on the noise power spectrum N_ℓ^i, characterized by σ_{N_ℓ}. These are estimated from two independent sets of $N_{\mathrm{sim}} = 100$ simulations, and the total uncertainty is obtained as their quadrature sum:

$$\sigma_\ell = \sqrt{\sigma_{C_\ell}^2 + \sigma_{N_\ell}^2}, \tag{13}$$

where σ_{C_ℓ} is the standard deviation of the corrected power spectra across sky-plus-noise realizations, and σ_{N_ℓ} is the standard deviation of the noise power spectra across noise-only realizations. Note that in the case of cross-spectra, the noise term N_ℓ^i does not appear in Equation 4 (since $\delta_{ij} = 0$), so $\sigma_{N_\ell} = 0$ and the total uncertainty reduces to $\sigma_\ell = \sigma_{C_\ell}$.

4.4.1 Sky signal simulations

Realistic simulated polarization sky maps for each frequency band and experiment are generated using the Python Sky Model (PySM) package[7] (Thorne et al., 2017). PySM is a widely used Python library for sky emission modelling in CMB analyses (e.g. Planck Collaboration et al. 2020c; Martire et al. 2022).

The simulation includes both polarized and intensity foreground components. Since this work focuses exclusively on polarization, only the polarized components enter the power spectrum analysis; the intensity components are included solely to produce realistic total-intensity maps but do

[7] https://pysm3.readthedocs.io/en/latest/

not contribute to the polarized signal and are not used in the fitting. The polarized foreground components are: synchrotron emission modelled as a power law with a spatially constant spectral index (**s1**); thermal dust emission described as a single-component modified blackbody (**d1**); and a lensed CMB polarization realization (**c1**), computed using the Taylens code (Næss & Louis, 2013) with unlensed C_ℓ produced by CAMB[8]. The intensity foreground components included are spinning dust (AME) modelled as a sum of two spinning dust populations (**a1**; Planck Collaboration 2016b), and free-free emission following the analytic Commander model (**f1**); these are included for completeness but play no role in the polarization analysis.

Simulated sky maps were generated independently for each of the 14 frequency bands using the PySM sky model. For each band, the simulated emission was expressed in thermodynamic CMB temperature units and convolved in harmonic space with the corresponding instrumental effective beam window function. The final maps were produced at a HEALPix resolution of $N_{\text{side}} = 512$. No bandpass integration was performed, each map was generated at the band-centre frequency.

4.4.2 Noise simulations

Noise simulations are produced independently for each frequency band and experiment, reflecting the distinct noise properties of each instrument.

QUIJOTE-MFI

The noise in the QUIJOTE-MFI maps is spatially correlated and anisotropic, and the two channels within each MFI horn exhibit significant noise correlations both in intensity ($\sim$ 60–80%) and in polarization ($\sim$ 20%; Rubiño-Martín et al. 2023). The QUIJOTE noise simulations used in this work are those described in Rubiño-Martín et al. (2023), which account for the measured anisotropy, spatial correlations, and inter-channel covariance structure.

WMAP

WMAP data are not significantly affected by $1/f$ noise, so the noise is well described by a per-pixel Gaussian model. Following Bennett et al. (2013), $N_{\text{sim}} = 100$ noise realization maps are generated as independent Gaussian random fields with zero mean and pixel-dependent standard deviation. The per-pixel noise level for polarization is:

$$\sigma_{\text{pix}} = \frac{\sigma_0}{\sqrt{N_{\text{obs, eff}}}}, \tag{14}$$

where σ_0 is the band-level noise sensitivity (tabulated in the WMAP data release) and $N_{\text{obs, eff}}$ is the effective number of observations per pixel, which accounts for the covariance between the Q and U Stokes parameters stored in the WMAP maps. Specifically, for the Q component:

$$N^{Q}_{\text{obs, eff}} = N^{Q}_{\text{obs}} - \frac{(N^{QU}_{\text{obs}})^2}{N^{U}_{\text{obs}}}, \tag{15}$$

and analogously for U, where N^{Q}_{obs}, N^{U}_{obs}, and N^{QU}_{obs} are the diagonal and off-diagonal elements of the per-pixel observation count matrix given by WMAP for each map and frequency band. This correction ensures that the effective noise per pixel correctly reflects the independent information content of Q and U after accounting for their covariance. The Q and U noise maps are drawn independently using their respective σ_{pix}.

Planck

For Planck LFI and HFI bands (PR3 data), $N_{\text{sim}} = 100$ per-band noise simulations are downloaded from the Planck Legacy Archive (Planck Collaboration et al., 2020c). These simulations are generated using the MCMC-based noise estimation procedures described in the respective data processing papers for LFI (Planck Collaboration, 2016a) and HFI (Planck Collaboration et al., 2020a), which characterize the noise power spectrum including contributions from the $1/f$ noise

[8] https://www.camb.info/

component of the detectors. Each simulation map is processed identically to the data in the analysis pipeline.

4.5. Foreground emission model

The measured power spectra over the frequency range 11-353 GHz are dominated by two Galactic emission components in polarization: synchrotron radiation and thermal dust emission (Planck Collaboration, 2016b). A cross-correlation term between the two components is included to account for any spatial correlation along the line of sight. This modelling approach is similar to that followed by Choi & Page (2015) and Hensley et al. (2022). The total model power spectrum between frequencies ν_i and ν_j we will use in this work is given by Equation 16.

$$C_\ell(\nu_i, \nu_j) = C_\ell^{\mathrm{sync}}(\nu_i, \nu_j) + C_\ell^{\mathrm{dust}}(\nu_i, \nu_j) + C_\ell^{\mathrm{corr}}(\nu_i, \nu_j). \tag{16}$$

Here C_ℓ^{sync} is the synchrotron power spectrum, modelled as a power law in both frequency and multipole space (Section 4.5.1). C_ℓ^{dust} is the thermal dust power spectrum, modelled via a modified blackbody SED (Section 4.5.2); and C_ℓ^{corr} is the synchrotron-dust cross-correlation term, which accounts for any spatial correlation between the two components (Section 4.5.3).

4.5.1 Synchrotron emission

Synchrotron radiation is the dominant polarized foreground at frequencies below ~70 GHz. Its angular power spectrum is well described by a power law in ℓ (Choi & Page, 2015; Martire et al., 2022). Defining the synchrotron frequency scaling as

$$S_s(\nu) = \left(\frac{\nu}{\nu_{\mathrm{ref}}}\right)^{\beta_s}, \tag{17}$$

the synchrotron power spectrum is:

$$C_\ell^{\mathrm{sync}}(\nu_i, \nu_j) = A_s \left(\frac{\ell}{\ell_{\mathrm{ref}}}\right)^{\alpha_s} S_s(\nu_i)\, S_s(\nu_j), \tag{18}$$

where A_s is the synchrotron amplitude at the reference multipole $\ell_{\mathrm{ref}} = 80$ and reference frequency $\nu_{\mathrm{ref}} = 23$ GHz, α_s is the multipole spectral index, and β_s is the frequency spectral index in Rayleigh-Jeans units.

4.5.2 Thermal dust emission

Thermal dust emission dominates polarized foreground emission above ~70 GHz, with a SED following a MBB law (Planck Collaboration et al., 2020b). In Rayleigh-Jeans brightness temperature units, the MBB scaling relative to the dust reference frequency $\nu_0 = 353$ GHz is:

$$S_d(\nu) = \left(\frac{\nu}{\nu_0}\right)^{\beta_d} \frac{B_\nu(\nu, T_d)}{B_\nu(\nu_0, T_d)}, \tag{19}$$

where B_ν is the Planck function and $T_d = 19.6$ K is the dust temperature, fixed in accordance with Planck measurements (Planck Collaboration, 2016b). The dust model power spectrum is:

$$C_\ell^{\mathrm{dust}}(\nu_i, \nu_j) = A_d \left(\frac{\ell}{\ell_{\mathrm{ref}}}\right)^{\alpha_d} S_d(\nu_i)\, S_d(\nu_j), \tag{20}$$

where A_d is the dust amplitude at $\ell_{\mathrm{ref}} = 80$ and $\nu_0 = 353$ GHz, α_d is the dust angular spectral index, and β_d is the MBB frequency spectral index.

4.5.3 Synchrotron-dust cross-correlation

A non-zero spatial correlation between synchrotron and thermal dust emission can arise when the GMF aligns the polarization angle of both components along the line of sight (Choi & Page, 2015). Following Hensley et al. (2022), the cross-correlation term is modelled as:

$$C_\ell^{\mathrm{corr}}(\nu_i, \nu_j) = \rho\sqrt{A_s A_d}\left[S_s(\nu_i)\, S_d(\nu_j) + S_s(\nu_j)\, S_d(\nu_i)\right]\left(\frac{\ell}{\ell_{\mathrm{ref}}}\right)^{(\alpha_s+\alpha_d)/2}, \tag{21}$$

where $\rho \in [-1, 1]$ is the synchrotron-dust correlation coefficient. When colour corrections are applied, each per-band scaling factor is divided by its corresponding colour correction polynomial, as described in Section 4.3.4.

4.5.4 CMB subtraction

Before fitting the foreground model, the CMB contribution is subtracted from the observed power spectra. We use the best-fit theoretical ΛCDM power spectrum from Planck 2018 (Planck Collaboration, 2020), obtained from the combined Plik TTTEEE likelihood with low-multipole (lowE) and gravitational lensing constraints. Both C_ℓ^{EE} and C_ℓ^{BB} are subtracted; for BB only the lensing contribution is included, since primordial tensor B-modes have not yet been detected. The CMB spectrum is interpolated to the effective multipoles of the binning scheme before subtraction.

4.6. Bayesian inference and MCMC fitting

We adopt a Bayesian framework to constrain the foreground model parameters from the measured power spectra. The full parameter vector for the power-law fitting is $\boldsymbol{\theta} = \{A_s, \alpha_s, \beta_s, A_d, \alpha_d, \beta_d, \rho\}$, where the EE and BB modes are fitted separately in this work. The posterior distribution of the parameters $\boldsymbol{\theta}$ is

$$p(\boldsymbol{\theta} \mid \mathbf{D}) \propto \mathcal{L}(\mathbf{D} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta}), \tag{22}$$

with $\mathcal{L}(\mathbf{D} \mid \boldsymbol{\theta})$ the likelihood of the data $\mathbf{D}$ given the parameters, and $p(\boldsymbol{\theta})$ the prior. The posterior is thus proportional to the product of the likelihood and the prior. The normalization constant (the Bayesian evidence) does not depend on the parameters $\boldsymbol{\theta}$ and is therefore irrelevant for parameter estimation.

4.6.1 Likelihood

The likelihood is constructed assuming that residuals between the measured and model power spectra are Gaussian distributed. This approximation is well justified in the regime of partial sky coverage and moderate multipoles ($\ell < 200$), where the central limit theorem ensures that the distribution of bandpower estimates is close to Gaussian, and has been widely adopted in foreground power spectrum analyses at similar angular scales (Choi & Page, 2015; Martire et al., 2022; Rubiño-Martín et al., 2023). The data vector $\mathbf{D}$ consists of the measured spectra $\hat{C}_\ell(\nu_i, \nu_j)$ concatenated over all frequency pairs and multipole bins, with corresponding uncertainties $\sigma_\ell(\nu_i, \nu_j)$ estimated from Monte Carlo realizations (Section 4.4.). The log-likelihood is then:

$$\ln \mathcal{L} = -\frac{1}{2}\sum_{i,j,\ell}\frac{[\hat{C}_\ell(\nu_i, \nu_j) - C_\ell(\nu_i, \nu_j; \boldsymbol{\theta})]^2}{\sigma_\ell^2(\nu_i, \nu_j)}. \tag{23}$$

4.6.2 Priors

Two fitting approaches are considered in this work: a power-law fit, in which all nine multipole bins are fitted simultaneously by a global power-law model, and a bin-to-bin fit, in which each multipole bin is fitted independently. Both approaches are described in detail in Section 4.7.. In both cases, uniform (flat) priors are adopted within physically motivated bounds:

$$A_s \geq 0, \quad A_d \geq 0, \quad \beta_s \leq 0, \quad \beta_d \geq 0, \quad \rho \in [-1,\, 1],$$

For the bin-to-bin analysis, an additional Gaussian prior on the synchrotron frequency spectral index is included to regularize the fit in multipole bins where the signal-to-noise ratio in the low-frequency channels is too low to adequately constrain the synchrotron SED, which can otherwise lead to spurious results for β_s and, consequently, for β_d (Planck Collaboration et al., 2020b):

$$\beta_s \sim \mathcal{N}(-3.1,\, 0.30^2), \tag{24}$$

consistent with the values found in published WMAP and Planck foreground analyses (Gold et al., 2011; Planck Collaboration et al., 2016). This approach has also been used in various works as in Planck Collaboration et al. (2020b) or Choi & Page (2015).

4.6.3 Sampler

Posterior sampling is performed with the EMCEE affine-invariant ensemble sampler (Foreman-Mackey et al., 2013). Convergence is assessed for all fits using a combination of standard diagnostics: the integrated autocorrelation time $\hat{\tau}$ is monitored to ensure the post-burn-in chain is sufficiently long; the effective sample size (ESS $= N_{\rm post}/\hat{\tau}$) is required to exceed 1000 per parameter; the mean acceptance fraction across walkers is verified to lie within the range recommended for the affine-invariant sampler; and the reduced chi-squared χ^2_{red} at the posterior maximum-likelihood point is inspected to confirm that the model provides an adequate fit and that the uncertainties are well calibrated.

4.7. Fitting approaches

Two complementary fitting approaches are applied to the corrected power spectra over the multipole range $20 \leq \ell \leq 200$, using data from all 14 frequency bands described in Section 3. The baseline analysis includes both auto- and cross-spectra, yielding 105 independent spectra.

Moreover, a complementary study restricted to cross-spectra only, which avoids any noise bias in the auto-spectra, is presented in Appendix A; and a study fitting in the multipole range $30 \leq \ell \leq 200$ suggested in Rubiño-Martín et al. (2023) is presented in Appendix B.

4.7.1 Power-law

In this approach all nine multipole bins are fitted simultaneously by the global power-law foreground model of Section 4.5.. The MCMC samples the full parameter vector $\boldsymbol{\theta} = \{A_s, \alpha_s, \beta_s, A_d, \alpha_d, \beta_d, \rho\}$ over all frequencies and multipoles. EE and BB modes are fitted separately.

4.7.2 Bin-to-bin

In this approach each multipole bin is fitted independently, without imposing any global power-law dependence on ℓ. The per-bin parameter vector is $\boldsymbol{\theta}_{\rm bin} = \{A_s, \beta_s, A_d, \beta_d, \rho\}$. The multipole spectral indices $(\alpha_s,\, \alpha_d)$ are not fitted in a single-bin analysis. This approach enables us to examine how the fitted parameters vary with the effective multipole, $\ell_{\rm eff}$, revealing any scale-dependent behavior or local deviations from the global power-law assumption.

5 RESULTS

In this section we present the results of the foreground power spectrum analysis described in Section 4. The analysis is performed over the multipole range $20 \leq \ell \leq 200$ using the combined 14-band of QUIJOTE, WMAP and Planck dataset. Two complementary fitting approaches are applied: a global power-law fit described in Section 4.7.1, and an independent bin-to-bin fit described in Section 4.7.2. For each approach, results are obtained for three Galactic masks defined by latitude cuts at $|b| > 10°$, $15°$, and $20°$ (Section 4.2.), allowing us to assess the robustness of the derived foreground parameters with respect to Galactic plane contamination. In all cases, the EE and BB polarization modes are fitted separately.

The foreground model and the Bayesian inference framework are described in Sections 4.5. and 4.6., respectively. This analysis follows approaches similar to those of Planck Collaboration et al. (2020b), Choi & Page (2015), Hensley et al. (2022), and Martire et al. (2022). The baseline results presented here include both auto- and cross-spectra (105 spectra in total). A complementary analysis restricted to cross-spectra only, which avoids any noise bias contribution from the auto-spectra, is presented in Appendix A.

The multipole range adopted here starts at $\ell = 20$, rather than the $\ell \geq 30$ range used in several comparable studies (Rubiño-Martín et al., 2023; Martire et al., 2022). The choice is motivated by the behaviour of the BB-mode power-law fit, as the BB signal is intrinsically fainter than EE and falls steeply with multipole, the first bin ($\ell_{\rm eff} = 29$) carries a large fraction of the constraining power on the angular spectral index α_s. When this bin is excluded, the remaining bins at smaller angular scales provide insufficient leverage to anchor the power-law slope, leading to steeper and poorly constrained values of α_s. Including the first bin resolves this degeneracy and yields well-constrained, physically consistent results for the BB mode. This effect is demonstrated in Appendix B, where results obtained over the range $30 \leq \ell \leq 200$ are presented for comparison.

5.1. Power-law fitting

The global power-law model (Section 4.7.1) is fitted simultaneously over all nine multipole bins and all 105 frequency spectra. The posterior constraints on the synchrotron amplitude A_s, multipole spectral index α_s, frequency spectral index β_s, dust amplitude A_d, multipole spectral index α_d, frequency spectral index β_d, and synchrotron-dust correlation coefficient ρ are reported as the median and 68% credible intervals. In order to quantify the goodness of the fit, in Appendix C, we present a table including all the reduced χ^2 values obtained for all power-law fittings of this work.

Results are obtained separately for EE and BB polarization modes, and for two dataset combinations: WMAP+Planck (WP+Pl) and QUIJOTE+WMAP+Planck (QJ+WP+Pl). These combinations are used to quantify the impact of including QUIJOTE-MFI data in the analysis.

Our best-fit results, obtained from the marginalized posterior distributions for each parameter, and for the three Galactic masks are presented in Table 1. Figures 3 and 4 show the corresponding best-fit synchrotron and thermal dust models, respectively, overlaid on the measured frequency power spectra for the $|b| > 10°$, 15° and 20° masks.

Table 1: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and synchrotron-dust correlation coefficient ρ, from a global power-law fit to the EE and BB polarization auto- and cross-spectra of the WP+Pl and QJ+WP+Pl datasets over the multipole range $20 \leq \ell \leq 200$, for the three Galactic masks.

Mask	Dataset	Mode	A_s [$10^{-3}\,\mu$K^2]	α_s	β_s	A_d [$10^{-3}\,\mu$K^2]	α_d	β_d	ρ
$\lvert b\rvert > 10°$	WP+Pl	EE	$6.687^{+0.421}_{-0.411}$	$-3.710^{+0.063}_{-0.065}$	$-3.017^{+0.041}_{-0.042}$	$3.217^{+0.011}_{-0.011}$	$-2.546^{+0.004}_{-0.004}$	$1.530^{+0.005}_{-0.005}$	$0.094^{+0.007}_{-0.007}$
		BB	$1.895^{+0.342}_{-0.317}$	$-3.258^{+0.172}_{-0.187}$	$-3.195^{+0.140}_{-0.145}$	$2.359^{+0.007}_{-0.007}$	$-2.203^{+0.004}_{-0.004}$	$1.530^{+0.005}_{-0.005}$	$0.132^{+0.013}_{-0.013}$
	QJ+WP+Pl	EE	$7.398^{+0.260}_{-0.256}$	$-3.619^{+0.037}_{-0.038}$	$-3.085^{+0.016}_{-0.016}$	$3.216^{+0.011}_{-0.011}$	$-2.547^{+0.004}_{-0.004}$	$1.530^{+0.005}_{-0.005}$	$0.096^{+0.005}_{-0.005}$
		BB	$1.657^{+0.186}_{-0.180}$	$-3.334^{+0.112}_{-0.121}$	$-2.999^{+0.052}_{-0.052}$	$2.359^{+0.007}_{-0.007}$	$-2.203^{+0.004}_{-0.004}$	$1.530^{+0.005}_{-0.005}$	$0.111^{+0.010}_{-0.010}$
$\lvert b\rvert > 15°$	WP+Pl	EE	$5.395^{+0.425}_{-0.415}$	$-3.874^{+0.078}_{-0.082}$	$-3.042^{+0.047}_{-0.048}$	$2.093^{+0.010}_{-0.010}$	$-2.534^{+0.006}_{-0.006}$	$1.552^{+0.007}_{-0.007}$	$0.080^{+0.008}_{-0.008}$
		BB	$1.723^{+0.376}_{-0.342}$	$-3.284^{+0.207}_{-0.229}$	$-3.314^{+0.175}_{-0.185}$	$1.378^{+0.006}_{-0.006}$	$-2.317^{+0.006}_{-0.006}$	$1.530^{+0.007}_{-0.007}$	$0.128^{+0.016}_{-0.015}$
	QJ+WP+Pl	EE	$5.990^{+0.260}_{-0.255}$	$-3.767^{+0.045}_{-0.046}$	$-3.079^{+0.017}_{-0.017}$	$2.092^{+0.010}_{-0.009}$	$-2.535^{+0.006}_{-0.006}$	$1.551^{+0.007}_{-0.007}$	$0.068^{+0.006}_{-0.006}$
		BB	$1.367^{+0.198}_{-0.183}$	$-3.410^{+0.141}_{-0.150}$	$-3.017^{+0.061}_{-0.061}$	$1.379^{+0.006}_{-0.006}$	$-2.317^{+0.006}_{-0.006}$	$1.531^{+0.007}_{-0.007}$	$0.116^{+0.013}_{-0.012}$
$\lvert b\rvert > 20°$	WP+Pl	EE	$5.948^{+0.451}_{-0.439}$	$-3.741^{+0.076}_{-0.080}$	$-3.140^{+0.054}_{-0.055}$	$1.439^{+0.009}_{-0.009}$	$-2.479^{+0.008}_{-0.008}$	$1.487^{+0.008}_{-0.008}$	$0.030^{+0.009}_{-0.009}$
		BB	$1.436^{+0.374}_{-0.336}$	$-3.476^{+0.243}_{-0.272}$	$-3.456^{+0.186}_{-0.198}$	$0.927^{+0.005}_{-0.005}$	$-2.237^{+0.007}_{-0.007}$	$1.532^{+0.009}_{-0.009}$	$0.162^{+0.019}_{-0.018}$
	QJ+WP+Pl	EE	$5.880^{+0.258}_{-0.253}$	$-3.732^{+0.046}_{-0.047}$	$-3.113^{+0.019}_{-0.019}$	$1.439^{+0.009}_{-0.009}$	$-2.478^{+0.008}_{-0.008}$	$1.486^{+0.008}_{-0.008}$	$0.023^{+0.007}_{-0.007}$
		BB	$1.347^{+0.208}_{-0.198}$	$-3.371^{+0.151}_{-0.164}$	$-2.984^{+0.065}_{-0.065}$	$0.927^{+0.005}_{-0.005}$	$-2.238^{+0.007}_{-0.007}$	$1.533^{+0.009}_{-0.009}$	$0.121^{+0.015}_{-0.015}$

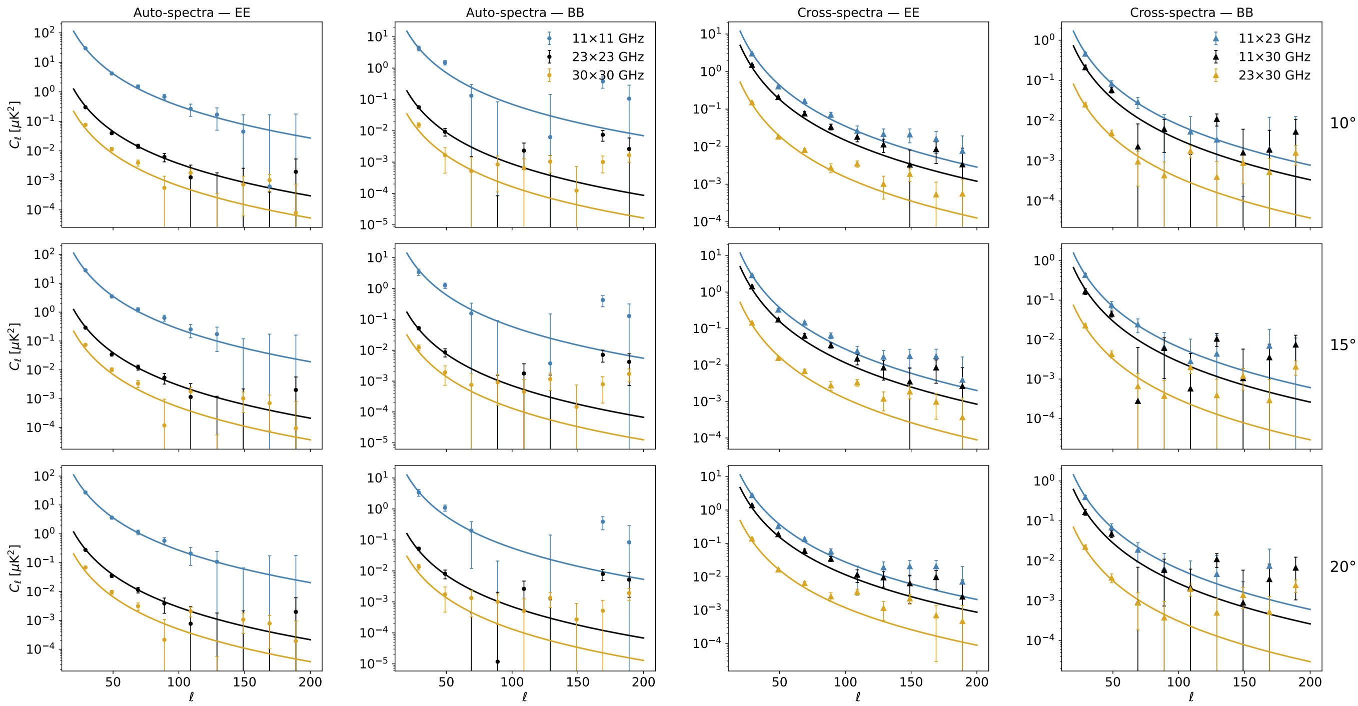

Figure 3: Measured auto- and cross-power spectra in the low-frequency bands (11, 23 and 30 GHz) for three Galactic masks. Each row corresponds to a mask (top to bottom: $|b| > 10°$, $15°$, $20°$) and the four columns show, from left to right, auto-spectra EE, auto-spectra BB, cross-spectra EE and cross-spectra BB. Markers are binned band powers C_ℓ with 1σ uncertainties (circles for auto-spectra, and triangles for cross-spectra). Solid lines show the best-fit full model from Equation 16 evaluated for each band pair.

The global power-law model provides a reasonable first-order description of the measured spectra over the fitted multipole range, as illustrated in Figures 3 and 4. In particular, the cross-spectra are generally well reproduced in both the synchrotron- and dust-dominated bands. For the auto-spectra, individual multipole bins occasionally deviate from the best-fit curve by more than 1σ, most noticeably at intermediate multipoles where the residual noise contributions differ between band pairs. These deviations are more pronounced for auto- than for cross-spectra, consistent with the fact that auto-spectra include an additional noise-bias term whose estimation and subtraction can introduce extra scatter. Overall, the fit captures the main frequency and multipole dependences of the data, but the reduced χ^2 values reported in Appendix C show that the agreement is not uniformly good across masks and modes. When the fit is restricted to cross-spectra only (Appendix A), the $\chi^2_{\rm red}$ values improve, supporting the interpretation that residual mismodelling of the auto-spectrum noise bias contributes to this scatter.

5.1.1 Synchrotron amplitude A_s

The synchrotron amplitude A_s is defined as the value of the power spectrum $C_\ell^{\rm sync}$ at the reference multipole $\ell_{\rm ref} = 80$ and at the reference frequency $\nu_{\rm ref} = 23\,$GHz (the WMAP K-band channel), as given by Equation 18. It is presented in Table 1 in units of $10^{-3}\,\mu$K^2 and is constrained separately for the EE and BB polarization modes.

The EE amplitude is consistently larger than the BB amplitude across all three masks and both dataset combinations, reflecting the well-established dominance of the E-mode over the B-mode in synchrotron polarization (Krachmalnicoff et al., 2018; Martire et al., 2022). For the baseline QJ+WP+Pl dataset, the EE amplitude ranges from $A_s^{EE} \approx 6.7 \cdot 10^{-3}\,\mu$K^2 for the $|b| > 10°$ mask to $A_s^{EE} \approx 5.9 \cdot 10^{-3}\,\mu$K^2 for the $|b| > 20°$ mask, reflecting the decreasing contribution of low-latitude emission as the Galactic plane emission is masked.

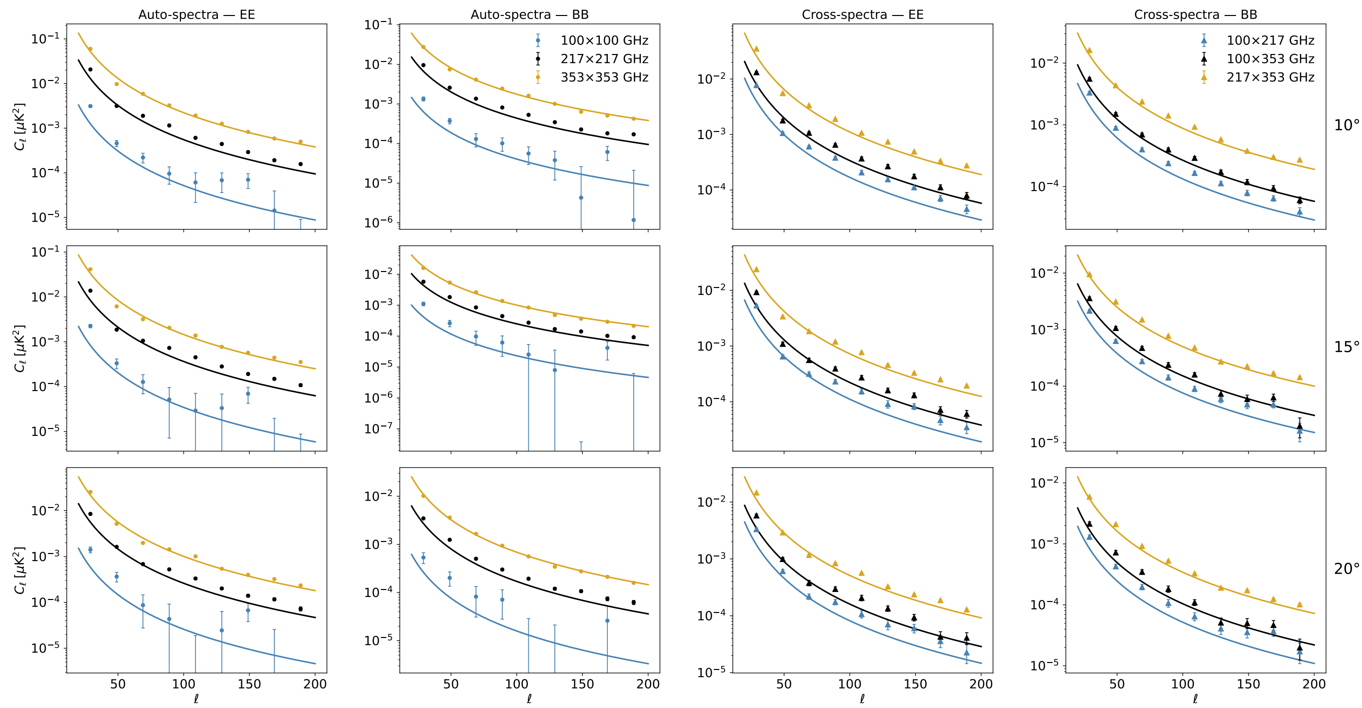

Figure 4: Measured auto- and cross-power spectra in the high-frequency bands (100, 217 and 353 GHz) for three Galactic masks. Each row corresponds to a mask (top to bottom: $|b| > 10°$, $15°$, $20°$) and the four columns show, from left to right, auto-spectra EE, auto-spectra BB, cross-spectra EE and cross-spectra BB. Markers are binned band powers C_ℓ with 1σ uncertainties (circles for auto-spectra, and triangles for cross-spectra). Solid lines show the best-fit full model from Equation 16 evaluated for each band pair.

The synchrotron B-to-E ratio $r \equiv A_s^{BB}/A_s^{EE}$ provides a direct measure of the relative power in the two polarization modes. For the WMAP and Planck dataset we find $r = 0.283 \pm 0.052$, 0.319 ± 0.071, and 0.241 ± 0.062 for the $|b| > 10°$, $15°$, and $20°$ masks, respectively. Adding QUIJOTE-MFI data tightens the constraints considerably and yields $r = 0.224 \pm 0.026$, 0.228 ± 0.033, and 0.229 ± 0.036 for the same three masks. The three values are mutually consistent within 1σ, indicating that the ratio is stable with respect to the choice of Galactic cut. The QJ+WP+Pl estimates are also in good agreement with the values 0.22 ± 0.02 reported by Martire et al. (2022) using WMAP and Planck data, and with the value 0.26 ± 0.08 obtained by Rubiño-Martín et al. (2023) using the QUIJOTE-MFI 11 GHz map over the range $30 < \ell < 200$ and the same $10°$ and $20°$ masks as we use in this work. These ratios are significantly lower than the values of ~ 0.5 found by Krachmalnicoff et al. (2018) using S-PASS data at 2.3 GHz, a discrepancy commonly attributed to Faraday rotation of the polarization angle at low frequencies and mid-Galactic latitudes (Fuskeland et al., 2021b). Compared to thermal dust, for which the B-to-E ratio is typically ~ 0.5 (Planck Collaboration et al., 2020b), synchrotron emission shows a markedly lower and more latitude-stable ratio of ~ 0.2.

The inclusion of the QUIJOTE-MFI data leads to a significant reduction in the uncertainty on A_s for both modes. For EE, the 1σ uncertainty is reduced by approximately a factor of 1.6 for all masks. For the BB mode, where the synchrotron signal is fainter and the constraints from WMAP and Planck alone are limited, the improvement is similarly notable, with uncertainties reduced by a factor of 1.8. This shows the key role of QUIJOTE-MFI in anchoring the low-frequency synchrotron amplitude, particularly for the BB mode.

5.1.2 Synchrotron angular spectral index α_s

The angular spectral index α_s parametrizes the multipole-dependence of the synchrotron power spectrum as a power law about the pivot $\ell_{\rm ref} = 80$ (Section 4.5.1). Negative values indicate that the power falls with increasing ℓ, as expected for diffuse Galactic emission dominated by large angular scales.

For the QJ+WP+Pl dataset, the EE spectral index takes values of $\alpha_s^{EE} \approx -3.7$ across all three masks, indicating a steep, large-scale-dominated spectrum. The BB spectral index is less tightly constrained, yielding $\alpha_s^{BB} \approx -3.3$ to -3.4 with uncertainties of ~ 0.12-0.16, also consistent across masks. The QUIJOTE-MFI data reduces the uncertainty on α_s^{EE} by roughly a factor of ~ 1.7 and on α_s^{BB} by a factor of ~ 1.5 with respect to WP+Pl alone, primarily by breaking the degeneracy between the angular and frequency dependences of the synchrotron model.

Although the BB values are nominally less steep than EE by $\Delta\alpha \approx 0.3$-0.4 across all masks and dataset combinations, the difference is only significant at about the $\sim 2\sigma$ level because the BB uncertainties are much larger. For that reason, the current data do not provide robust evidence for a genuine difference between the EE and BB angular slopes. This becomes even clearer when the baseline fit is compared with the $30 \leq \ell \leq 200$ case discussed in Appendix B. Once the first multipole bin ($\ell_{\rm eff} = 29$) is removed, the trend reverses and the BB slope becomes steeper than the EE slope for all three masks (e.g., $\alpha_s^{BB} \approx -4.1$ versus $\alpha_s^{EE} \approx -2.9$ for $|b| > 10°$), but with very large BB uncertainties. The fact that the sign of the $\alpha_s^{EE} - \alpha_s^{BB}$ difference changes with the fitting range shows that the relative slope is not yet robustly constrained and remains sensitive to the low-ℓ binning.

This same dependence on the fitted multipole range also helps explain the comparison with previous works. The baseline EE values found here ($\alpha_s^{EE} \approx -3.6$ to -3.8) are steeper than the -2.95 ± 0.07 reported by Martire et al. (2022), who used WMAP and Planck data over a sky fraction of 0.4 and a fitting interval of $30 \leq \ell \leq 300$. Our results are also steeper than the -3.16 ± 0.03 obtained by Krachmalnicoff et al. (2018) using S-PASS at 2.3 GHz over $20 \leq \ell \leq 140$ and a sky fraction of 0.22. However, in Appendix B, when the first bin is excluded, the EE slopes flatten to $\alpha_s^{EE} \approx -2.7$ to -3.0, bringing them much closer to the reference values. This suggests that part of the discrepancy is driven by the extra leverage introduced by the $\ell_{\rm eff} = 29$ bin in our baseline analysis, while any remaining disagreement with S-PASS may also reflect the different sky regions covered by the two experiments. Notably, Krachmalnicoff et al. (2018) also reported steep EE slopes when the $\ell_{\rm eff} = 29$ bin was included, obtaining $\alpha_s^{EE} = -3.32 \pm 0.07$ and a flatter $\alpha_s^{BB} = -3.03 \pm 0.12$ for a $|b| > 40°$ mask, a pattern consistent with the behaviour found in the present work.

To further investigate the sensitivity of α_s to the binning choice, we performed a set of auxiliary single-band fits using the QUIJOTE 11 GHz, WMAP 23 GHz, and Planck 30 GHz maps independently. In these tests, each band is fitted to a simple power law in multipole space with only amplitude A and slope α as free parameters (i.e., without the frequency spectral index β_s, since no frequency combination is involved).

Two model variants were considered. In the first case, the data were fitted with a pure power-law model. In the second case, an additional constant term (independent of the multipole) was included in the model to account for possible mismodelings of the noise and contributions from point radio sources. We performed these tests using two different binning schemes, these are: $\Delta\ell = 10$ and $\Delta\ell = 20$.

In all cases, the fitted constant (offset) term was consistent with zero for all three bands, indicating that such an offset is not required in the baseline foreground fitting.

With $\Delta\ell = 10$ binning, the slope estimates remain broadly consistent with $\alpha_s \approx -3$ for both polarization modes across all three bands, in good agreement with the global power-law results obtained in Rubiño-Martín et al. (2023) and Martire et al. (2022). With $\Delta\ell = 20$ binning the fitting is more sensitive to whether the lowest-ℓ bin is included. When the first bin is excluded ($\ell_{\rm eff} = 29$), the EE slope tends toward flatter values near -3, while the BB mode is too faint to be well constrained, yielding large uncertainties and unreliable slope estimates, particularly for WMAP and Planck, explaining the results presented in Appendix B for the multipole spectral index. When the first bin is included, the EE slope steepens, consistent with what is seen in Table 1, but the BB fit improves notably as the high-signal first bin provides the leverage needed to anchor the power-law slope, leading to well-constrained values of α_s^{BB} near -3. This confirms that the steeper EE slopes in the baseline analysis are a consequence of the additional low-ℓ leverage, and that including this bin is essential for obtaining reliable BB constraints with the current data and binning scheme.

In this context, it is useful to consider the reduced χ^2 values reported in Table 11 (Appendix C). For the global power-law fit, the $\chi^2_{\rm red}$ values are in several cases significantly above unity, particularly for the EE mode at low Galactic latitude cuts, suggesting that the true multipole dependence of the synchrotron power spectrum is not perfectly captured by a single power law over the range $20 \leq \ell \leq 200$. In contrast, the bin-to-bin results (Section 5.2.) yield $\chi^2_{\rm red}$ values close to unity for most bins and masks, confirming that the model is flexible enough to describe the data at each individual scale. This discrepancy between the global and per-bin goodness of fit implies that the angular spectral index α_s derived from the power-law fit should not be over-interpreted, as it represents an effective average slope over the full multipole range, but the underlying spectrum may deviate from a strict power law, and its precise value is consequently sensitive to the fitting range and binning choice.

5.1.3 Synchrotron frequency spectral index β_s

The frequency spectral index β_s governs the power-law scaling of the synchrotron SED in Rayleigh-Jeans brightness temperature units, $T_{\rm RJ} \propto \nu^{\beta_s}$, and is directly related to the energy distribution of the cosmic-ray electrons responsible for synchrotron radiation (Rybicki & Lightman, 1979; Padovani et al., 2021). Of all the parameters in the foreground model, β_s is the one most directly sensitive to the frequency coverage of the dataset, because its constraint comes almost entirely from the lever arm between the lowest and highest frequency channels at which synchrotron emission is detectable. This makes β_s the parameter for which the inclusion of the QUIJOTE-MFI bands produces the most notable improvement.

Without QUIJOTE, the WP+Pl dataset yields EE spectral indices in the range $\beta_s^{EE} \approx -3.0$ to -3.1 across the three masks, with uncertainties of 0.04-0.06.The corresponding BB estimates are nominally steeper by $\sim$ 0.1-0.3, but consistent with the EE values within 1-2σ given BB uncertainties of $\sim$ 0.14-0.19. These relatively large errors reflect the limited frequency baseline of WMAP and Planck alone, for which the synchrotron signal is not sufficiently strong or well separated from other components to tightly constrain the spectral index.

Adding the two QUIJOTE-MFI bands extends the synchrotron baseline down to 11 GHz, making a significant improvement in the fitting. For the QJ+WP+Pl combination the EE uncertainties decrease by a factor of 2.6-2.9 depending on mask, and the BB uncertainties by a factor of 2.8-3.0, making β_s the parameter with the largest error reduction brought by QUIJOTE. The resulting QJ+WP+Pl posteriors converge tightly around $\beta_s \approx -3.0$ to -3.1 for both polarization modes, with EE errors of ~ 0.02 and BB errors of order ~ 0.06. Across all three masks, the EE and BB estimates are mutually consistent within $\sim 2\sigma$, providing no statistically significant evidence for a difference between the two polarization modes.

These values are in excellent agreement with the canonical synchrotron spectral index, $\beta_s \approx -3.1$, reported in previous studies. Fuskeland et al. (2014) found $\beta_s = -3.12 \pm 0.04$ from an analysis of WMAP polarization data in real space, while Martire et al. (2022) obtained $\beta_s^{EE} = -3.00 \pm 0.10$ and $\beta_s^{BB} = -3.05 \pm 0.36$ using a similar multi-frequency power spectrum approach combining WMAP and Planck data. Our QJ+WP+Pl EE estimates ($\beta_s^{EE} \approx -3.08$ to -3.11) are consistent with both studies within 1σ. For the BB-mode spectral indices, we obtain $\beta_s^{BB} \approx -2.98$ to -3.02. These values are compatible with the results of Martire et al. (2022), while providing a clear improvement in the constraints on this parameter for the BB polarization mode. Furthermore, Fuskeland et al. (2021b) measured a steeper value, $\beta_s = -3.22 \pm 0.08$, using Faraday-corrected S-PASS data combined with WMAP. This measurement is also consistent with our results at the $\sim 2\sigma$ level.

5.1.4 Synchrotron-dust spatial correlation ρ

The correlation coefficient ρ quantifies the degree of spatial correlation between the synchrotron and thermal dust polarized emission along the line of sight (Section 4.5.3; Choi & Page, 2015). A positive ρ indicates that the two foreground components share a common polarization structure, which is physically expected when the GMF aligns the polarization directions of both components over large angular scales (Choi & Page, 2015). This parameter is constrained jointly with the other foreground parameters in the global power-law fit and is bounded by $\rho \in [-1, 1]$.

For both dataset combinations and all three masks, we recover small but significantly non-zero positive values of ρ, confirming a weak spatial correlation between the two foreground components. For the QJ+WP+Pl dataset, the EE correlation coefficient decreases with increasing Galactic latitude cut, from $\rho^{EE} = 0.096 \pm 0.005$ for $|b| > 10°$ to $\rho^{EE} = 0.023 \pm 0.007$ for $|b| > 20°$. This behaviour indicates that, as the mask becomes more restrictive and the retained sky is increasingly dominated by high-latitude regions, the spatial correlation between the two components decreases. The BB correlation coefficient shows a different trend. The parameter ρ^{BB} remains relatively stable across masks, taking values of ~ 0.11-0.12 for the QJ+WP+Pl dataset, and does not exhibit the pronounced decline observed for the EE mode.

The difference between the EE and BB correlation coefficients increases with the Galactic cut. For the $|b| > 10°$ mask, the two modes are consistent at the $\sim 1.3\sigma$ level. The discrepancy grows to $\sim 3.4\sigma$ for $|b| > 15°$ and reaches $\sim 5.9\sigma$ for $|b| > 20°$, providing evidence that the two polarization modes trace spatially distinct correlation structures at high Galactic latitudes. Choi & Page (2015) and Planck Collaboration et al. (2020b) reported a similar decrease in the EE correlation as the masking becomes more restrictive in a bin-to-bin analysis. However, their estimates present significant deviations across bins and considerably large uncertainties due to the limited synchrotron signal in their data, preventing firm conclusions about this behaviour.

The overall magnitude and mode-dependent behaviour of ρ are consistent with previous studies. Planck Collaboration et al. (2020b) also reported low but positive correlations in their joint synchrotron-dust power spectrum analysis, finding ρ values consistent with ours at the ~ 0.1 level. Krachmalnicoff et al. (2018), using S-PASS at 2.3 GHz combined with Planck 353 GHz data, found per-bin values of ρ that are generally positive and range from near zero up to ~ 0.40 in the most inclusive masks at the lowest multipoles, but that cluster around 0.1-0.2 across most bins, multipoles, and sky fractions for both polarization modes. This behaviour is in good agreement with our global estimates of $\rho \sim 0.1$-0.12, given that these values represent averages over the full multipole range. Our results confirm that the synchrotron-dust correlation, while non-negligible, remains at the level of $\rho \sim 0.1$ or below for most of the sky outside the Galactic plane, and exhibits different behaviour in the E- and B-modes.

5.1.5 Thermal dust parameters

The thermal dust component is parametrized by the amplitude A_d at $\ell_{\rm ref} = 80$ and $\nu_0 = 353\,$GHz, the angular spectral index α_d, and the modified-blackbody frequency spectral index β_d (Section 4.5.2). Because the dust signal is anchored by the Planck HFI bands (100-353 GHz), which are present in both dataset combinations, the inclusion of QUIJOTE data has virtually no effect on the dust parameters. The WP+Pl and QJ+WP+Pl posteriors are identical for all three dust parameters. This is expected, since the QUIJOTE bands at 11 and 17 GHz carry negligible dust emission, and confirms that the synchrotron and dust parameters are well separated by the fit.

The EE dust amplitude decreases monotonically from $A_d^{EE} \approx 3.2 \cdot 10^{-3}\,\mu{\rm K}^2$ for the $|b| > 10°$ mask to $A_d^{EE} \approx 1.4 \cdot 10^{-3}\,\mu{\rm K}^2$ for the $|b| > 20°$ mask, reflecting the decreasing density of dust-emitting material at higher Galactic latitudes. The BB amplitude follows the same trend, decreasing from $A_d^{BB} \approx 2.4 \cdot 10^{-3}$ to $0.9 \cdot 10^{-3}\,\mu{\rm K}^2$, and is consistently smaller than EE. The dust B-to-E ratio A_d^{BB}/A_d^{EE} ranges from 0.734 ± 0.003 ($|b| > 10°$) to 0.644 ± 0.005 ($|b| > 20°$), in broad agreement with the ratio of ~ 0.5-0.6 found by Planck Collaboration et al. (2020b) and Planck Collaboration et al. (2016). Our values are slightly higher, which may reflect the different multipole ranges and sky fractions used.

The EE angular index is $\alpha_d^{EE} \approx -2.5$ for all three masks, with the $|b| > 20°$ mask yielding a slightly flatter value of -2.478 ± 0.008. The BB values are consistently less steep, with $\alpha_d^{BB} \approx -2.2$ to -2.3, indicating that the dust BB power spectrum falls off less steeply with multipole than EE. This EE-BB difference is highly significant, exceeding 5σ for all masks. Over a similar multipole range, Planck Collaboration et al. (2020b) reported $\alpha_d^{EE} = -2.42 \pm 0.02$ and $\alpha_d^{BB} = -2.54 \pm 0.02$, with BB marginally steeper than EE, in contrast to the trend found in our results. Planck Collaboration et al. (2016) reported for both EE-and BB-modes $\alpha_d \approx -2.42 \pm 0.02$ for high-latitude sky at 353 GHz over $40 < \ell < 600$. Our EE values are close to this estimate, while the BB values are slightly flatter, which may reflect the different multipole ranges and sky fractions used in the two analyses.

The MBB spectral index β_d is remarkably stable across all masks, dataset combinations, and polarization modes, with values clustering tightly around $\beta_d \approx 1.53$. The EE estimates range from 1.486 ± 0.008 ($|b| > 20°$) to 1.552 ± 0.007 ($|b| > 15°$), while the BB values span 1.530 ± 0.005 to 1.533 ± 0.009. The EE and BB values are mutually consistent within $\sim 1\sigma$ for all masks except $|b| > 20°$, where the EE value is slightly lower. These values are in excellent agreement with the Planck measurements of $\beta_d = 1.53 \pm 0.02$ from polarization data (Planck Collaboration et al., 2020b) and with the intensity-based estimates of $\beta_d \approx 1.51$-1.59 from Planck Collaboration et al. (2015). The tight consistency of β_d across masks and modes confirms that the dust SED is well characterized by a single modified blackbody with $T_d = 19.6\,\mathrm{K}$ over the sky fractions and multipole range considered in this work.

5.2. Bin-to-bin fitting

In the bin-to-bin approach (see Section 4.7.2), each of the nine multipole bins is fitted independently, without imposing any global power-law dependence on ℓ. The per-bin parameter vector is $\boldsymbol{\theta}_{\text{bin}} = \{A_s, \beta_s, A_d, \beta_d, \rho\}$, and results are obtained separately for EE and BB modes. This approach allows the identification of scale-dependent trends and potential deviations from the global power-law behaviour found in Section 5.1.. The Gaussian prior $\beta_s \sim \mathcal{N}(-3.1, 0.3^2)$ (Section 4.6.2) is applied to all bins, but only becomes relevant in those where the low-frequency signal-to-noise ratio is insufficient to independently constrain the synchrotron SED, in the remaining bins the data drive the posterior well within the prior width. This prior is anchored by the global power-law constraints derived in Section 5.1.3, where the QJ+WP+Pl posteriors yield β_s uncertainties of ~ 0.02 for EE and ~ 0.06 for BB, being an order of magnitude narrower than the prior width of 0.3, confirming that the posterior in well-constrained bins is dominated by the data rather than the prior.

In this section we will account all auto- and cross-spectra of the QUIJOTE, WMAP and Planck dataset for the fit. Posterior constraints (median and 68% credible intervals) on A_s, β_s, A_d, β_d, and ρ are reported as a function of the effective multipole ℓ_{eff} for each bin, including the reduced χ^2 values obtained for each bin of the fitting. The full numerical results for all three masks are listed in Tables 2, 3 and 4. The corresponding cross-spectra-only results are provided in Appendix A.

5.2.1 Amplitude ratios

Figure 5 shows the B-to-E amplitude ratios A_s^{BB}/A_s^{EE} and A_d^{BB}/A_d^{EE} as a function of ℓ_{eff} for each mask. The synchrotron amplitude ratio scatters around the reference line $A_s^{BB}/A_s^{EE} = 0.2$ with no systematic trend with scale or mask, in good agreement with the global estimates of ~ 0.22-0.23 (Section 5.1.1) and with the results of Rubiño-Martín et al. (2023) and Martire et al. (2022). The larger scatter at high ℓ is expected due to the decreasing signal-to-noise ratio of the synchrotron B-mode amplitude in individual bins.

The dust ratio is likewise approximately stable, clustering around 0.6-0.7 and consistent with the global values of ~ 0.64-0.73, in agreement with results found by Planck Collaboration et al. (2020b), which show that at low multipoles the measured B-to-E power ratio is in the range of about 0.5-1. However, these values are consistently higher than the ~ 0.5 reported by Choi & Page (2015) and Planck Collaboration et al. (2016). Notably, Figure 5 shows a clear dependence on the

Galactic mask, as the more restrictive cuts yield a lower dust ratio, suggesting that residual large-scale Galactic emission may bias the amplitude ratio upward in the most inclusive sky fractions.

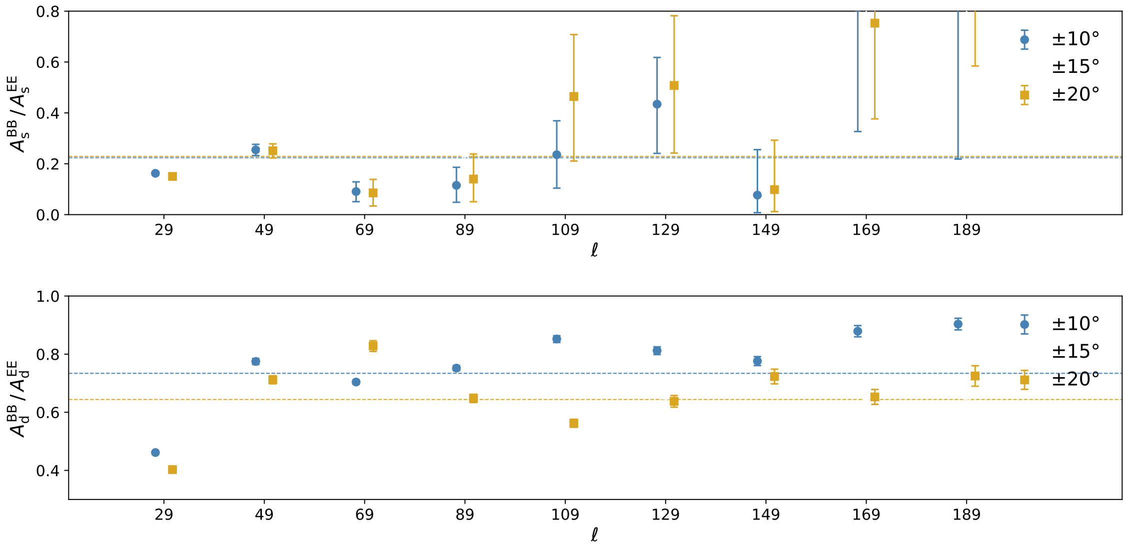

Figure 5: Synchrotron and dust B-to-E amplitude ratios A_s^{BB}/A_s^{EE} (top panel) and A_d^{BB}/A_d^{EE} (bottom panel), for the three Galactic cuts masks from the bin-to-bin analysis (Tables 2, 3 and 4) using the QJ+WP+Pl dataset. The dashed line marks the $A_s^{BB}/A_s^{EE} = 0.25$ and $A_d^{BB}/A_d^{EE} = 0.70$ values as a reference.

5.2.2 Synchrotron frequency spectral index β_s

The per-bin β_s posteriors are shown in Figure 6 and are broadly consistent with the global power-law estimates (Section 5.1.3). For the well-constrained EE bins at low ℓ, the values cluster around $\beta_s^{EE} \approx -3.09$ to -3.24 across all three masks, in good agreement with the global QJ+WP+Pl results of $\beta_s^{EE} \approx -3.08$ to -3.11. Individual bins at intermediate and high multipoles have uncertainties of order 0.2-0.5, as the signal-to-noise ratio decreases and the Gaussian prior $\beta_s \sim \mathcal{N}(-3.1, 0.3^2)$ progressively governs the posterior. For BB, the first bin ($\ell_{\rm eff} = 29$) is well constrained with $\beta_s^{BB} \approx -3.03$ to -3.10 across masks, consistent with the global estimates. At higher multipoles the synchrotron B-mode signal is too faint to independently constrain the spectral index, and the posterior widens considerably. No significant scale dependence of β_s is detected in either polarization mode. The scatter of the estimates around -3.1 is consistent with statistical noise and the effect of the prior in low signal-to-noise bins. These results confirm the findings of Section 5.1.3 and are consistent with Martire et al. (2022), as well as with Choi & Page (2015) and Planck Collaboration et al. (2020b), who adopted a similar bin-to-bin approach including a Gaussian prior on the synchrotron spectral index.

5.2.3 Dust spectral index β_d

The per-bin β_d posteriors are shown in Figure 7. As expected from the dominance of the Planck HFI bands in constraining the dust SED, β_d is well recovered in each bin and is remarkably stable across both multipoles and masks. EE values cluster around $\beta_d^{EE} \approx 1.5$ to 1.6 across individual bins, scattering around the canonical 1.53 with no systematic trend with $\ell_{\rm eff}$. Most bins are consistent with the global EE estimates (~ 1.53, Section 5.1.5) within 1-2σ. The BB estimates are even more stable, clustering around $\beta_d^{BB} \approx 1.5$, consistent with the global values of ~ 1.53. The tight consistency across bins and masks confirms that the modified-blackbody approximation with fixed $T_d = 19.6\,$K provides a valid description of the dust SED over the full multipole range $20 \leq \ell \leq 200$, in agreement with Planck Collaboration et al. (2020b) and Planck Collaboration et al. (2015).

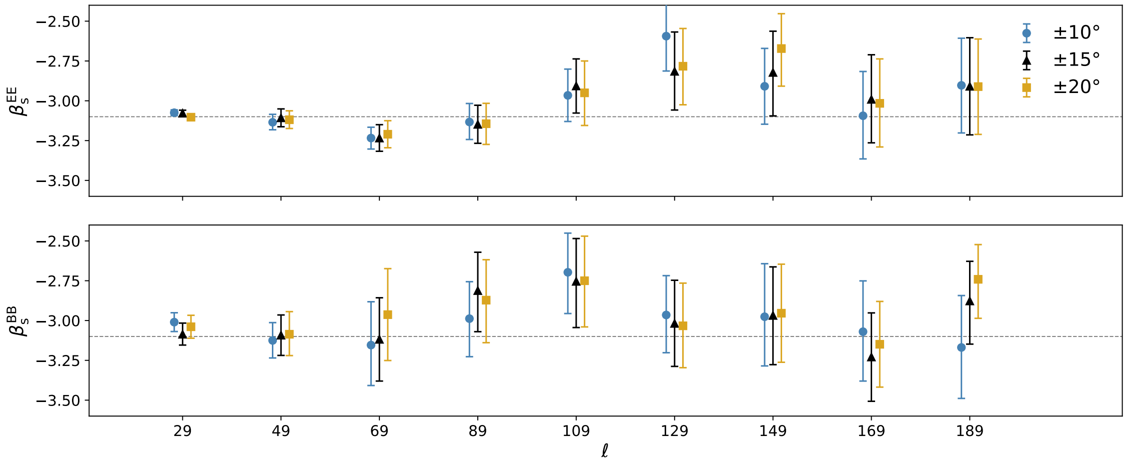

Figure 6: Synchrotron spectral index β_s for E-modes (top panel) and B-modes (bottom panel), for the three Galactic cuts masks from the bin-to-bin analysis (Tables 2, 3 and 4) using the QJ+WP+Pl dataset. The dashed line marks the $\beta_s = -3.1$ value as a reference.

5.2.4 Synchrotron-dust spatial correlation ρ

The per-bin ρ posteriors are shown in Figure 8. For EE, the most constrained estimates are obtained at low ℓ, where the values are small and positive, generally in the range $\rho^{EE} \sim 0.09$-0.10 depending on the mask, consistent with the trend of decreasing correlation with increasing Galactic latitude cut found in the global fit (Section 5.1.4). At higher multipoles the EE estimates become noisier, but remain broadly consistent with small positive values at the $\sim 10\%$ level, in agreement with Planck Collaboration et al. (2020b).

For BB, the per-bin estimates are noisier, especially at small scale bins, where the noise dominates the data. Notably, a handful of high-ℓ BB bins yield small negative values, but none exceed 1.5σ significance. Al larger scales ($\ell \leq 130$), the values scatter around ~ 0.1 at all multipoles, consistent with the stable $\rho^{BB} \sim 0.11$-0.12 found in the global fit. No significant trend with $\ell_{\rm eff}$ is present in either mode. The overall results are consistent with the findings of Choi & Page (2015), who reported per-bin values of $\rho \sim 0.2$ from WMAP 9-year data, and with Krachmalnicoff et al. (2018), who found values clustering around 0.1-0.2 across most bins and sky fractions. Our results confirm that the synchrotron-dust spatial correlation is a real but weak signal at the level of $\rho \sim 0.1$-0.2, most prominent in EE at large angular scales and in the most inclusive sky fractions.

In summary, the bin-to-bin analysis confirms the parameter values obtained in the global power-law fit and provides no evidence for significant scale-dependent deviations in any of the foreground parameters over the multipole range $20 \leq \ell \leq 200$. The spectral indices β_s and β_d are consistent with their global values in all well-constrained bins, the synchrotron-dust correlation ρ remains small and positive, and the amplitude ratios are stable with ℓ. Together, these results support the validity of the global power-law parametrization adopted in Section 5.1. as an adequate description of the foreground emission over the sky fractions and multipole range considered in this work.

5.3. Goodness of fit

The reduced χ^2 values for each bin and mask are listed in the respective bin-to-bin tables. For both EE and BB modes, most bins yield $\chi^2_{\rm red} \approx 1.0$-$1.6$ across all masks, indicating a good agreement between the data and the per-bin model. This should be contrasted with the global power-law fit (Table 11, Appendix C), which produces substantially larger values in most cases. For the QJ+WP+Pl combination over $20 \leq \ell \leq 200$, the EE reduced χ^2 reaches 7.90, 6.85, and 2.85 for the $|b| > 10°$, 15°, and 20° masks, respectively, while the BB values are 2.50, 1.84, and 1.60. The improvement in the bin-to-bin case confirms that the per-bin model, which imposes no global

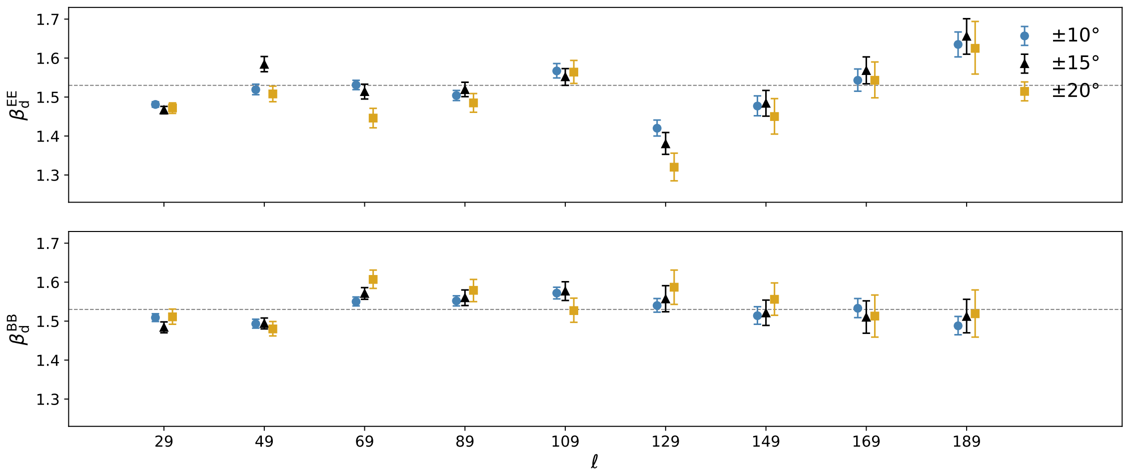

Figure 7: Dust spectral index β_d for E-modes (top panel) and B-modes (bottom panel), for the three Galactic cuts masks from the bin-to-bin analysis (Tables 2, 3 and 4) using the QJ+WP+Pl dataset. The dashed line marks the $\beta_d = 1.53$ value as a reference.

ℓ-dependence, is able to describe the data at each individual scale, whereas the single power-law model is unable to fully account for the multipole structure of the synchrotron emission over the full range $20 \leq \ell \leq 200$ (see Section 5.1.2).

The power-law $\chi^2_{\rm red}$ values improve noticeably both with more restrictive masking and when the first multipole bin is excluded. Restricting the fit to $30 \leq \ell \leq 200$ reduces the EE values to 3.27, 1.95, and 1.69 for the three masks, approaching acceptability. The BB $\chi^2_{\rm red}$ values remain similar across the two multipole intervals, although the posterior parameter constraints differ significantly between these fitting ranges.

This shows that the poor EE goodness of fit in the baseline analysis is largely driven by the tension introduced by the $\ell_{\rm eff} = 29$ bin, which carries significant constraining power but may not be well described by the same power law that fits the higher-ℓ bins.

The bin-to-bin $\chi^2_{\rm red}$ also tends to decrease with more restrictive masking, although this trend must be interpreted cautiously, as at high Galactic latitudes the data are noisier, which widens the per-bin uncertainties and therefore lowers the $\chi^2_{\rm red}$ rather than reflecting a better model description.

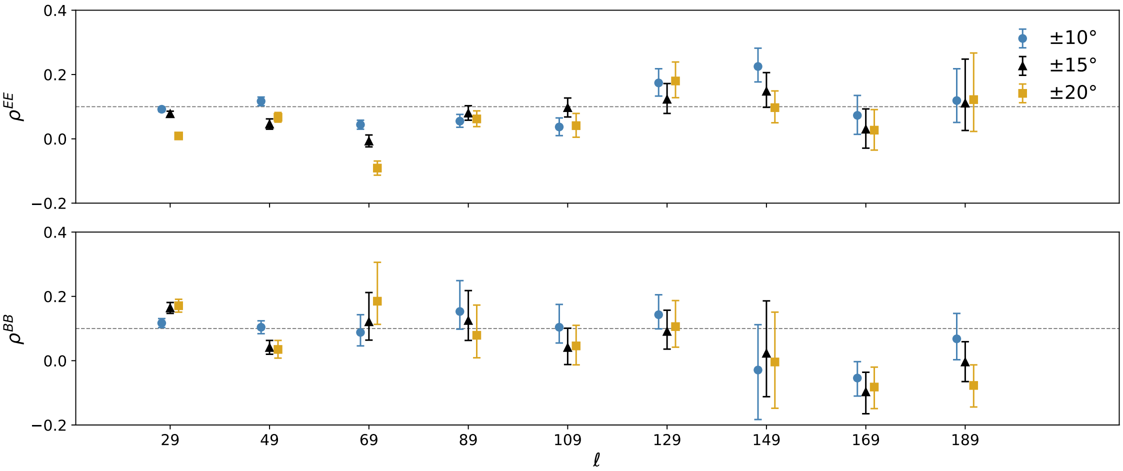

Figure 8: Synchrotron-dust spatial correlation ρ for E-modes (top panel) and B-modes (bottom panel), for the three Galactic cuts masks from the bin-to-bin analysis (Tables 2, 3 and 4) using the QJ+WP+Pl dataset.The dashed line marks the $\rho = 0.1$ value as a reference.

Table 2: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and the synchrotron-dust correlation coefficient ρ. The parameters are fitted independently in each multipole bin from the EE (top) and BB (bottom) polarization auto- and cross-spectra of the 14-band QJ+WP+Pl dataset, using the $|b| > 10°$ Galactic mask.

ℓ range	$\ell_{\rm eff}$	$A_{\rm s}^{\rm EE}$ [$10^{-3}\,\mu$K^2]	$\beta_{\rm s}^{\rm EE}$	$A_{\rm d}^{\rm EE}$ [$10^{-3}\,\mu$K^2]	$\beta_{\rm d}^{\rm EE}$	$\rho^{\rm EE}$	$\chi^2_{\rm red}$
20–39	29.0	$294.505^{+3.295}_{-3.397}$	$-3.075^{+0.018}_{-0.018}$	$49.086^{+0.270}_{-0.261}$	$1.481^{+0.007}_{-0.007}$	$0.092^{+0.007}_{-0.007}$	1.564
40–59	49.0	$37.590^{+1.238}_{-1.322}$	$-3.135^{+0.050}_{-0.047}$	$7.887^{+0.077}_{-0.084}$	$1.519^{+0.014}_{-0.013}$	$0.117^{+0.013}_{-0.014}$	1.251
60–79	69.0	$13.055^{+0.753}_{-0.759}$	$-3.234^{+0.068}_{-0.069}$	$4.798^{+0.040}_{-0.041}$	$1.531^{+0.012}_{-0.012}$	$0.044^{+0.014}_{-0.014}$	1.169
80–99	89.0	$6.238^{+0.606}_{-0.639}$	$-3.133^{+0.116}_{-0.110}$	$2.673^{+0.022}_{-0.023}$	$1.504^{+0.013}_{-0.013}$	$0.055^{+0.021}_{-0.019}$	1.433
100–119	109.0	$3.519^{+0.507}_{-0.523}$	$-2.966^{+0.165}_{-0.164}$	$1.555^{+0.016}_{-0.017}$	$1.567^{+0.019}_{-0.018}$	$0.037^{+0.028}_{-0.027}$	1.084
120–139	129.0	$2.535^{+0.477}_{-0.500}$	$-2.594^{+0.230}_{-0.219}$	$1.017^{+0.013}_{-0.013}$	$1.420^{+0.021}_{-0.020}$	$0.174^{+0.044}_{-0.041}$	1.495
140–159	149.0	$1.980^{+0.583}_{-0.593}$	$-2.909^{+0.238}_{-0.238}$	$0.679^{+0.010}_{-0.010}$	$1.477^{+0.026}_{-0.025}$	$0.225^{+0.057}_{-0.048}$	0.955
160–179	169.0	$1.299^{+0.524}_{-0.502}$	$-3.094^{+0.278}_{-0.271}$	$0.479^{+0.008}_{-0.008}$	$1.543^{+0.029}_{-0.028}$	$0.073^{+0.062}_{-0.059}$	1.002
180–199	189.0	$1.070^{+0.658}_{-0.617}$	$-2.903^{+0.296}_{-0.299}$	$0.404^{+0.007}_{-0.007}$	$1.635^{+0.032}_{-0.032}$	$0.119^{+0.099}_{-0.068}$	1.212
ℓ range	$\ell_{\rm eff}$	$A_{\rm s}^{\rm BB}$ [$10^{-3}\,\mu$K^2]	$\beta_{\rm s}^{\rm BB}$	$A_{\rm d}^{\rm BB}$ [$10^{-3}\,\mu$K^2]	$\beta_{\rm d}^{\rm BB}$	$\rho^{\rm BB}$	$\chi^2_{\rm red}$
20–39	29.0	$47.841^{+1.913}_{-1.914}$	$-3.010^{+0.059}_{-0.059}$	$22.650^{+0.173}_{-0.177}$	$1.509^{+0.010}_{-0.010}$	$0.117^{+0.014}_{-0.014}$	1.389
40–59	49.0	$9.572^{+0.753}_{-0.772}$	$-3.125^{+0.112}_{-0.110}$	$6.111^{+0.052}_{-0.052}$	$1.493^{+0.012}_{-0.011}$	$0.104^{+0.020}_{-0.020}$	1.394
60–79	69.0	$1.185^{+0.492}_{-0.516}$	$-3.154^{+0.272}_{-0.254}$	$3.378^{+0.023}_{-0.024}$	$1.550^{+0.012}_{-0.011}$	$0.088^{+0.055}_{-0.042}$	1.562
80–99	89.0	$0.718^{+0.438}_{-0.408}$	$-2.988^{+0.232}_{-0.239}$	$2.010^{+0.016}_{-0.016}$	$1.552^{+0.013}_{-0.013}$	$0.153^{+0.096}_{-0.055}$	1.360
100–119	109.0	$0.829^{+0.454}_{-0.446}$	$-2.697^{+0.246}_{-0.259}$	$1.325^{+0.011}_{-0.012}$	$1.572^{+0.015}_{-0.015}$	$0.104^{+0.071}_{-0.049}$	1.156
120–139	129.0	$1.101^{+0.417}_{-0.440}$	$-2.965^{+0.247}_{-0.237}$	$0.826^{+0.008}_{-0.009}$	$1.540^{+0.018}_{-0.017}$	$0.143^{+0.062}_{-0.044}$	1.695
140–159	149.0	$0.152^{+0.351}_{-0.129}$	$-2.976^{+0.333}_{-0.309}$	$0.527^{+0.007}_{-0.007}$	$1.514^{+0.023}_{-0.022}$	$-0.029^{+0.141}_{-0.154}$	1.153
160–179	169.0	$1.057^{+0.486}_{-0.483}$	$-3.070^{+0.319}_{-0.310}$	$0.421^{+0.006}_{-0.006}$	$1.533^{+0.025}_{-0.024}$	$-0.054^{+0.051}_{-0.056}$	1.475
180–199	189.0	$1.095^{+0.653}_{-0.586}$	$-3.169^{+0.326}_{-0.320}$	$0.365^{+0.005}_{-0.005}$	$1.488^{+0.024}_{-0.023}$	$0.068^{+0.079}_{-0.065}$	1.646

Table 3: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and the synchrotron-dust correlation coefficient ρ. The parameters are fitted independently in each multipole bin from the EE (top) and BB (bottom) polarization auto- and cross-spectra of the 14-band QJ+WP+Pl dataset, using the $|b| > 15°$ Galactic mask.

ℓ range	$\ell_{\rm eff}$	$A_{\rm s}^{\rm EE}$ [$10^{-3}\,\mu$K^2]	$\beta_{\rm s}^{\rm EE}$	$A_{\rm d}^{\rm EE}$ [$10^{-3}\,\mu$K^2]	$\beta_{\rm d}^{\rm EE}$	$\rho^{\rm EE}$	$\chi^2_{\rm red}$
20–39	29.0	$276.355^{+3.451}_{-3.415}$	$-3.078^{+0.019}_{-0.019}$	$33.564^{+0.235}_{-0.242}$	$1.467^{+0.009}_{-0.010}$	$0.078^{+0.008}_{-0.008}$	2.025
40–59	49.0	$32.205^{+1.230}_{-1.292}$	$-3.107^{+0.056}_{-0.056}$	$4.943^{+0.069}_{-0.071}$	$1.584^{+0.020}_{-0.019}$	$0.046^{+0.016}_{-0.016}$	1.039
60–79	69.0	$10.944^{+0.757}_{-0.774}$	$-3.234^{+0.084}_{-0.083}$	$2.612^{+0.033}_{-0.034}$	$1.514^{+0.019}_{-0.019}$	$-0.007^{+0.019}_{-0.018}$	1.150
80–99	89.0	$5.791^{+0.602}_{-0.638}$	$-3.148^{+0.120}_{-0.119}$	$1.691^{+0.020}_{-0.021}$	$1.519^{+0.019}_{-0.018}$	$0.080^{+0.023}_{-0.022}$	1.418
100–119	109.0	$3.234^{+0.489}_{-0.500}$	$-2.907^{+0.170}_{-0.170}$	$1.118^{+0.014}_{-0.014}$	$1.552^{+0.021}_{-0.022}$	$0.097^{+0.030}_{-0.029}$	1.294
120–139	129.0	$2.109^{+0.515}_{-0.508}$	$-2.815^{+0.247}_{-0.243}$	$0.623^{+0.011}_{-0.011}$	$1.380^{+0.029}_{-0.027}$	$0.123^{+0.049}_{-0.044}$	1.390
140–159	149.0	$1.952^{+0.606}_{-0.609}$	$-2.822^{+0.259}_{-0.273}$	$0.461^{+0.009}_{-0.009}$	$1.484^{+0.033}_{-0.033}$	$0.148^{+0.058}_{-0.050}$	1.058
160–179	169.0	$1.552^{+0.564}_{-0.546}$	$-2.991^{+0.280}_{-0.273}$	$0.363^{+0.007}_{-0.007}$	$1.568^{+0.035}_{-0.034}$	$0.030^{+0.063}_{-0.059}$	0.932
180–199	189.0	$0.815^{+0.689}_{-0.580}$	$-2.909^{+0.305}_{-0.305}$	$0.288^{+0.007}_{-0.007}$	$1.656^{+0.045}_{-0.046}$	$0.111^{+0.137}_{-0.085}$	1.184
ℓ range	$\ell_{\rm eff}$	$A_{\rm s}^{\rm BB}$ [$10^{-3}\,\mu$K^2]	$\beta_{\rm s}^{\rm BB}$	$A_{\rm d}^{\rm BB}$ [$10^{-3}\,\mu$K^2]	$\beta_{\rm d}^{\rm BB}$	$\rho^{\rm BB}$	$\chi^2_{\rm red}$
20–39	29.0	$41.336^{+2.018}_{-2.036}$	$-3.086^{+0.070}_{-0.068}$	$13.185^{+0.145}_{-0.146}$	$1.484^{+0.014}_{-0.014}$	$0.164^{+0.017}_{-0.017}$	1.514
40–59	49.0	$8.650^{+0.794}_{-0.812}$	$-3.092^{+0.127}_{-0.127}$	$4.407^{+0.044}_{-0.046}$	$1.494^{+0.014}_{-0.014}$	$0.041^{+0.022}_{-0.021}$	1.672
60–79	69.0	$0.866^{+0.520}_{-0.506}$	$-3.118^{+0.261}_{-0.262}$	$2.148^{+0.019}_{-0.019}$	$1.571^{+0.015}_{-0.015}$	$0.121^{+0.091}_{-0.057}$	1.405
80–99	89.0	$0.832^{+0.487}_{-0.471}$	$-2.812^{+0.241}_{-0.258}$	$1.119^{+0.014}_{-0.014}$	$1.560^{+0.020}_{-0.020}$	$0.125^{+0.093}_{-0.062}$	1.172
100–119	109.0	$0.953^{+0.501}_{-0.506}$	$-2.754^{+0.269}_{-0.290}$	$0.688^{+0.009}_{-0.009}$	$1.577^{+0.024}_{-0.024}$	$0.041^{+0.060}_{-0.053}$	1.025
120–139	129.0	$1.012^{+0.441}_{-0.437}$	$-3.018^{+0.271}_{-0.270}$	$0.393^{+0.007}_{-0.007}$	$1.557^{+0.034}_{-0.033}$	$0.091^{+0.066}_{-0.055}$	1.524
140–159	149.0	$0.186^{+0.382}_{-0.156}$	$-2.968^{+0.305}_{-0.309}$	$0.308^{+0.005}_{-0.005}$	$1.521^{+0.033}_{-0.032}$	$0.023^{+0.163}_{-0.135}$	1.193
160–179	169.0	$1.104^{+0.471}_{-0.454}$	$-3.229^{+0.277}_{-0.278}$	$0.237^{+0.006}_{-0.005}$	$1.510^{+0.042}_{-0.041}$	$-0.097^{+0.061}_{-0.068}$	1.258
180–199	189.0	$1.859^{+0.642}_{-0.660}$	$-2.877^{+0.249}_{-0.271}$	$0.190^{+0.005}_{-0.005}$	$1.512^{+0.044}_{-0.042}$	$-0.004^{+0.063}_{-0.061}$	1.337

Table 4: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and the synchrotron-dust correlation coefficient ρ. The parameters are fitted independently in each multipole bin from the EE (top) and BB (bottom) polarization auto- and cross-spectra of the 14-band QJ+WP+Pl dataset, using the $|b| > 20°$ Galactic mask.

ℓ range	$\ell_{\rm eff}$	$A_{\rm s}^{\rm EE}$ [$10^{-3}\,\mu$K^2]	$\beta_{\rm s}^{\rm EE}$	$A_{\rm d}^{\rm EE}$ [$10^{-3}\,\mu$K^2]	$\beta_{\rm d}^{\rm EE}$	$\rho^{\rm EE}$	$\chi^2_{\rm red}$
20–39	29.0	$263.649^{+3.842}_{-3.803}$	$-3.103^{+0.021}_{-0.021}$	$20.685^{+0.229}_{-0.224}$	$1.472^{+0.013}_{-0.014}$	$0.009^{+0.010}_{-0.009}$	1.636
40–59	49.0	$32.168^{+1.267}_{-1.293}$	$-3.119^{+0.056}_{-0.055}$	$4.137^{+0.062}_{-0.061}$	$1.508^{+0.020}_{-0.020}$	$0.067^{+0.015}_{-0.015}$	1.526
60–79	69.0	$10.382^{+0.738}_{-0.721}$	$-3.210^{+0.085}_{-0.085}$	$1.624^{+0.029}_{-0.029}$	$1.446^{+0.025}_{-0.025}$	$-0.091^{+0.022}_{-0.022}$	1.096
80–99	89.0	$5.343^{+0.607}_{-0.608}$	$-3.144^{+0.128}_{-0.130}$	$1.178^{+0.018}_{-0.018}$	$1.485^{+0.024}_{-0.024}$	$0.062^{+0.025}_{-0.024}$	1.263
100–119	109.0	$2.461^{+0.513}_{-0.513}$	$-2.950^{+0.200}_{-0.205}$	$0.820^{+0.014}_{-0.014}$	$1.564^{+0.030}_{-0.029}$	$0.041^{+0.038}_{-0.036}$	1.189
120–139	129.0	$1.955^{+0.504}_{-0.498}$	$-2.783^{+0.237}_{-0.242}$	$0.436^{+0.010}_{-0.010}$	$1.320^{+0.036}_{-0.035}$	$0.180^{+0.059}_{-0.052}$	1.317
140–159	149.0	$2.391^{+0.584}_{-0.601}$	$-2.672^{+0.219}_{-0.236}$	$0.325^{+0.009}_{-0.009}$	$1.450^{+0.046}_{-0.045}$	$0.097^{+0.052}_{-0.047}$	1.069
160–179	169.0	$1.769^{+0.586}_{-0.569}$	$-3.016^{+0.279}_{-0.274}$	$0.265^{+0.007}_{-0.007}$	$1.543^{+0.047}_{-0.045}$	$0.027^{+0.064}_{-0.062}$	0.998
180–199	189.0	$0.829^{+0.681}_{-0.572}$	$-2.911^{+0.299}_{-0.300}$	$0.189^{+0.006}_{-0.006}$	$1.625^{+0.069}_{-0.066}$	$0.122^{+0.145}_{-0.099}$	1.077
ℓ range	$\ell_{\rm eff}$	$A_{\rm s}^{\rm BB}$ [$10^{-3}\,\mu$K^2]	$\beta_{\rm s}^{\rm BB}$	$A_{\rm d}^{\rm BB}$ [$10^{-3}\,\mu$K^2]	$\beta_{\rm d}^{\rm BB}$	$\rho^{\rm BB}$	$\chi^2_{\rm red}$
20–39	29.0	$39.584^{+1.976}_{-1.985}$	$-3.039^{+0.072}_{-0.072}$	$8.333^{+0.113}_{-0.113}$	$1.511^{+0.020}_{-0.019}$	$0.171^{+0.020}_{-0.020}$	1.804
40–59	49.0	$8.077^{+0.815}_{-0.851}$	$-3.086^{+0.142}_{-0.134}$	$2.945^{+0.036}_{-0.037}$	$1.480^{+0.019}_{-0.018}$	$0.035^{+0.028}_{-0.027}$	1.368
60–79	69.0	$0.888^{+0.547}_{-0.533}$	$-2.963^{+0.289}_{-0.288}$	$1.345^{+0.017}_{-0.019}$	$1.607^{+0.024}_{-0.023}$	$0.185^{+0.121}_{-0.072}$	1.464
80–99	89.0	$0.747^{+0.520}_{-0.469}$	$-2.872^{+0.254}_{-0.267}$	$0.763^{+0.012}_{-0.012}$	$1.579^{+0.028}_{-0.029}$	$0.079^{+0.094}_{-0.070}$	1.517
100–119	109.0	$1.143^{+0.550}_{-0.577}$	$-2.750^{+0.280}_{-0.290}$	$0.461^{+0.008}_{-0.008}$	$1.527^{+0.032}_{-0.030}$	$0.046^{+0.064}_{-0.059}$	1.016
120–139	129.0	$0.993^{+0.471}_{-0.454}$	$-3.033^{+0.268}_{-0.263}$	$0.278^{+0.006}_{-0.006}$	$1.587^{+0.044}_{-0.044}$	$0.106^{+0.081}_{-0.064}$	1.419
140–159	149.0	$0.235^{+0.461}_{-0.198}$	$-2.954^{+0.308}_{-0.308}$	$0.235^{+0.005}_{-0.005}$	$1.556^{+0.042}_{-0.041}$	$-0.004^{+0.155}_{-0.144}$	1.239
160–179	169.0	$1.332^{+0.527}_{-0.510}$	$-3.149^{+0.269}_{-0.269}$	$0.173^{+0.005}_{-0.005}$	$1.513^{+0.054}_{-0.054}$	$-0.082^{+0.062}_{-0.067}$	1.092
180–199	189.0	$2.067^{+0.669}_{-0.686}$	$-2.741^{+0.218}_{-0.245}$	$0.137^{+0.005}_{-0.005}$	$1.519^{+0.061}_{-0.060}$	$-0.077^{+0.064}_{-0.067}$	1.197

6 CONCLUSIONS

In this work we have presented a comprehensive characterization of the spectral properties of the two dominant polarized Galactic foregrounds, the synchrotron and thermal dust emission through a multi-frequency angular power spectrum analysis covering the frequency range from 11 to 353 GHz. The analysis combines the QUIJOTE-MFI DR1 maps at 11 and 17 GHz, all five WMAP 9-year frequency bands (23-94 GHz), and the seven polarization bands of the Planck PR3 dataset (28.4-353 GHz), forming a total of 105 auto- and cross-power spectra. We apply two complementary approaches by fitting the data to a global power-law model and performing an independent bin-to-bin fit. Each method is carried out for the EE and BB polarization modes separately and over three Galactic masks with latitude cuts at $|b| > 10°$, $15°$, and $20°$.

The global power-law fitting yields well-constrained posterior distributions for all foreground parameters. For the synchrotron component, the frequency spectral index obtained from the full QJ+WP+Pl dataset is $\beta_s^{EE} \approx -3.08$ to -3.11 and $\beta_s^{BB} \approx -2.98$ to -3.02, with uncertainties at the level of 0.02 (EE) and 0.06 (BB). These values are in excellent agreement with the canonical index of $\beta_s \approx -3.1$ reported in the literature (Fuskeland et al., 2014; Martire et al., 2022), and compatible at $\sim 2\sigma$ with the slightly steeper value of $\beta_s = -3.22 \pm 0.08$ obtained by Fuskeland et al. (2021b) using Faraday-corrected S-PASS data. The QUIJOTE-MFI bands provide the largest single improvement in the determination of the synchrotron spectral index. By extending the synchrotron frequency baseline down to 11 GHz, the inclusion of QUIJOTE data reduces the 1σ uncertainty on β_s by a factor of 2.6-3.0 with respect to the WMAP+Planck dataset alone. This highlights the importance of the QUIJOTE-MFI observations in breaking the degeneracy between the synchrotron amplitude and spectral index, making them an essential input for any foreground characterization aimed at supporting component separation for future CMB polarization experiments.

The synchrotron B-to-E amplitude ratio is one of the most robust results of this analysis.For the QJ+WP+Pl dataset, we obtain $r \equiv A_s^{BB}/A_s^{EE} \approx$ 0.22-0.23, consistent within 1σ across all masks. These values are in excellent agreement with $r = 0.22 \pm 0.02$ from Martire et al. (2022) and with $r = 0.26 \pm 0.08$ from Rubiño-Martín et al. (2023), and are significantly lower than the ~ 0.5 reported at 2.3 GHz by Krachmalnicoff et al. (2018) with S-PASS, a discrepancy commonly attributed to Faraday rotation of the polarization angle at low frequencies (Fuskeland et al., 2021b). For thermal dust we find $A_d^{BB}/A_d^{EE} \approx$ 0.64–0.73, broadly consistent with the ~ 0.5 reported by Planck Collaboration et al. (2020b) and Planck Collaboration et al. (2016). In contrast, the synchrotron emission exhibits a markedly lower and more latitude-stable E/B power asymmetry.

The angular spectral indices α_s of the synchrotron component deserve particular attention. In the baseline analysis over $20 \leq \ell \leq 200$, the EE mode yields $\alpha_s^{EE} \approx -3.6$ to -3.8 for all masks and the BB mode yields $\alpha_s^{BB} \approx -3.3$ to -3.4, both steeper than the values of $\alpha_s \approx -2.94$ reported by Martire et al. (2022) and $\alpha_s \approx -3.16$ reported by Krachmalnicoff et al. (2018). The Appendix B analysis over $30 \leq \ell \leq 200$ reveals that part of this steepening is driven by the additional low-ℓ leverage introduced by including the first multipole bin ($\ell_{\rm eff} = 29$). When this bin is removed, the EE slopes flatten to $\alpha_s^{EE} \approx -2.9$ to -3.0, in much better agreement with the other works. For the BB mode, excluding the first bin has the opposite and more pronounced effect, the slope steepens to $\alpha_s^{BB} \approx -3.6$ to -4.1 with significantly larger uncertainties, demonstrating that the first bin is indispensable for obtaining a reliable BB angular spectral index with the current binning scheme. These results show that the shape of the synchrotron angular power spectrum, particularly the BB mode, remains sensitive to the choice of multipole range. This sensitivity is further reflected in the goodness of fit of the global power-law model as the $\chi^2_{\rm red}$ values for the QJ+WP+Pl fit over $20 \leq \ell \leq 200$ are in several cases substantially above unity, reaching $\chi^2_{\rm red} = 7.90$ (EE) and 2.50 (BB) for the $|b| > 10°$ mask and improving to 2.85 and 1.60 for $|b| > 20°$. Excluding the first bin brings the EE values down to 3.27-1.69 depending on the mask, confirming that the dominant tension originates from the $\ell_{\rm eff} = 29$ bin. In contrast, the bin-to-bin fit yields $\chi^2_{\rm red}$ of 1.0-1.6 for most bins and masks, demonstrating that the per-bin model describes the data adequately at each individual scale. This comparison implies that the angular spectral index α_s from the power-law fit should not be over-interpreted, as it represents an effective average slope over the full multipole range, but the underlying spectrum may deviate from a strict power law, and its precise value is consequently sensitive to the fitting range and binning choice.

One of the key new results of this work is the characterization of the spatial cross-correlation between synchrotron and thermal dust polarized emission, encoded by the coefficient ρ. We consistently recover small but significantly non-zero positive values across all fitting configurations. For the QJ+WP+Pl dataset, the EE correlation decreases monotonically with the Galactic latitude cut, from $\rho^{EE} = 0.096 \pm 0.005$ ($|b| > 10°$) to $\rho^{EE} = 0.023 \pm 0.007$ ($|b| > 20°$). In contrast, ρ^{BB} is stable at $\sim$ 0.11-0.12 across all masks. The statistical significance of the EE-BB difference in ρ increases from $\sim 1.3\sigma$ at $|b| > 10°$ to $\sim 5.9\sigma$ at $|b| > 20°$, providing evidence that the two polarization modes trace spatially distinct magnetic field structures.

The overall values found in this work are consistent with those reported by Choi & Page (2015) ($\rho \sim 0.2$) and with the per-bin values clustering around 0.1-0.2 in Krachmalnicoff et al. (2018), and confirm that the synchrotron-dust cross-correlation, while weak, is a real and physically meaningful signal that should be included in foreground models.

The bin-to-bin analysis independently corroborates all of the above findings. The per-bin spectral indices β_s and β_d are consistent with the global power-law estimates throughout the full multipole range, with no evidence for significant scale-dependent deviations. The synchrotron-dust correlation ρ remains small and positive in all well-constrained bins, and the B-to-E amplitude ratios for both components are stable with ℓ. The global power-law parametrization is therefore validated as an adequate description of the foreground emission over the sky fractions and multipole range considered in this work, in agreement with Martire et al. (2022), and Planck Collaboration et al. (2020b).

The robustness of the results was further verified through two complementary analyses. The cross-spectra-only fit (Appendix A), which eliminates any contribution from noise bias in the auto-spectra, yields parameter estimates fully consistent with the baseline auto+cross results within 1-2σ for all parameters, masks, and polarization modes. The alternative fitting range $30 \leq \ell \leq 200$ (Appendix B) demonstrates that all foreground parameters except the synchrotron angular spectral index are stable to the choice of lower multipole limit, confirming that the sensitivity to the multipole range is essentially confined to the angular power-law slopes, and in particular to the synchrotron BB mode.

In summary, this work delivers a comprehensive and self-consistent characterization of polarized Galactic foreground emission over the Northern sky, making use of a wide frequency baseline in polarization by combining QUIJOTE, WMAP, and Planck data. The results provide tight constraints on the synchrotron and dust spectral properties and on their cross-correlation with the GMF, and are intended to serve as legacy foreground inputs for component separation in forthcoming CMB polarization experiments.

6.1. Future work

Several natural extensions of this work are foreseen and will be explored in future studies. For instance, the foreground model presented here accounts for synchrotron and thermal dust emission as the two dominant polarized Galactic components, and does not include AME. While AME is observed to have a very low polarization fraction (Génova-Santos et al., 2017; González-González et al., 2025), and its theoretical origin in spinning dust grains predicts negligible levels of intrinsic polarization (Draine et al., 2016; Herman, D. et al., 2023), its contribution to the total polarized sky at intermediate frequencies (20-70 GHz) is not yet fully characterized. An important direction for future work is to incorporate AME polarization into the foreground model, making use of the unique sensitivity of the QUIJOTE-MFI bands to this emission and complementing them with other forthcoming data. This extension would allow us to obtain upper limits or detections of the AME polarization fraction in a power-spectrum framework, and to assess whether any residual AME contribution biases the recovered synchrotron spectral parameters in the intermediate frequency range.

A second key direction is the study of the spatial variability of the synchrotron spectral index. The current analysis characterizes foreground properties over large sky fractions ($f_{\rm sky} \approx 0.23$-0.30), effectively averaging over a wide range of astrophysical environments along many different lines of sight. However, it is well established that the synchrotron spectral index β_s exhibits significant spatial variations across the sky, reflecting local changes in the cosmic-ray electron energy distribution and in the strength and geometry of the GMF (Krachmalnicoff et al., 2018; Rubiño-Martín et al., 2023). These variations are a source of foreground complexity that cannot be captured by a spatially uniform model and represent one of the main challenges for component separation in CMB polarization searches. A forthcoming analysis will extend the present work by carrying out the same multi-frequency power-spectrum fit over a set of small sky patches covering approximately 5-10% of the sky each.

This region-by-region approach will enable a direct mapping of the spatial dependence of β_s at the power-spectrum level, providing a study of the spectral properties of the synchrotron and dust emission across the sky, quantifying their amplitude, angular scale, and spatial variability, and offering observational constraints that are currently unavailable at frequencies above 23 GHz. The wide frequency baseline and the low-frequency data provided by the QUIJOTE-MFI bands make this analysis particularly well suited to detecting and quantifying β_s variations in the 10-100 GHz range, which is most critical for the component separation of future CMB B-mode experiments.

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A Analysis fitting only cross-spectra

This appendix presents the results of the global power-law and bin-to-bin fitting analyses when only cross-spectra are used, i.e. excluding all auto-spectra from the likelihood. This configuration eliminates any noise bias contribution to the fitted signal, since cross-spectra between independent detector splits are unaffected by the noise auto-correlation. It therefore serves as a robustness check on the baseline results presented in Section 5, and on the goodness of fit discussed in relation to Appendix C.

1.1. Power-law fitting

The cross-spectra-only posterior constraints over $20 \leq \ell \leq 200$ are reported in Table 5 and should be compared directly with the baseline auto+cross results in Table 1. The two sets of results are in excellent agreement. All parameter estimates are consistent within 1-2σ across all masks, dataset combinations, and polarization modes. The synchrotron amplitudes A_s, frequency spectral indices β_s, angular spectral indices α_s, and all dust parameters (A_d, α_d, β_d) show no statistically significant shift when the auto-spectra are removed. Similarly, the correlation coefficient ρ remains stable between the two configurations. This confirms that the baseline results are not driven by noise bias in the auto-spectra, and that the foreground model is robustly constrained by the cross-spectra alone.

The main observable difference is in the parameter uncertainties. Removing the auto-spectra reduces the number of spectra accounted for the fit from 105 to 91, which results in a systematic increase in the posterior widths. For the QJ+WP+Pl dataset, the 1σ uncertainties on A_s^{EE} and α_s^{EE} increase by roughly 10-15%, while the BB uncertainties are similarly affected. The increase is larger for β_s^{BB} in the WP+Pl case, where the auto-spectra contribute meaningfully to breaking the synchrotron-noise degeneracy at low frequencies. Overall, the constraining power remains largely unaffected.

The reduced χ^2 values (Table 11, Appendix C) improve, but not significantly when auto-spectra are not accounted. For the EE mode with QJ+WP+Pl, χ^2_{red} decreases by a factor $\sim$ 1.2-1.4 in the cross-only configuration. For the BB mode there is also an improvement, however, only by a factor $\sim$ 1.1-1.2. The EE values remain elevated even in the cross-only case, particularly for the $|b| > 10°$ mask, which suggests that the residual mismatch reflects genuine limitations of the single power-law model at low latitudes rather than noise misestimation in the auto-spectra. The BB χ^2_{red} values are consistently closer to unity in both configurations, indicating that the model captures the BB spectra more accurately, consistent with the simpler signal structure of the B modes, but also as a consequence of the larger uncertainties present in the BB power spectra.

Figures 9 and 10 show the best-fit models overlaid on the measured spectra for the synchrotron- and dust-dominated bands, respectively. The visual appearance is very similar to the baseline Figures 3 and 4, with the cross-spectra curves following the data equally well, as expected from the consistency of the fitted parameters.

1.2. Bin-to-bin fitting

The cross-spectra-only bin-to-bin results are reported in Tables 6, 7 and 8, and should be compared with the all-pairs results in Tables 2, 3 and 4 respectively. The two sets of results are in excellent agreement across all bins, masks, and polarization modes. The per-bin β_s posteriors cluster around -3.0 to -3.3 in both configurations, with no systematic shift when the auto-spectra are removed. The dust spectral index β_d remains stable around 1.53 in both EE and BB throughout the full multipole range. The correlation coefficient ρ shows the same qualitative behaviour as in the all-pairs case, presenting small and positive EE values at low ℓ decreasing with Galactic latitude cut, and noisier BB estimates consistent with zero to $\sim$ 0.1 at all scales. All per-bin estimates are consistent within 1-2σ between the two configurations.

Table 5: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and synchrotron-dust correlation coefficient ρ, from a global power-law fit to the EE and BB polarization cross-spectra of the WP+Pl and QJ+WP+Pl datasets over the multipole range $20 \leq \ell \leq 200$, for the three Galactic masks.

Mask	Dataset	Mode	$A_s\,[10^{-3}\,\mu\mathrm{K}^2]$	α_s	β_s	$A_d\,[10^{-3}\,\mu\mathrm{K}^2]$	α_d	β_d	ρ
$\lvert b\rvert > 10^\circ$	WP+Pl	EE	$7.196^{+0.572}_{-0.547}$	$-3.716^{+0.076}_{-0.079}$	$-3.177^{+0.065}_{-0.067}$	$3.178^{+0.019}_{-0.019}$	$-2.542^{+0.006}_{-0.006}$	$1.520^{+0.008}_{-0.008}$	$0.095^{+0.007}_{-0.007}$
		BB	$2.273^{+0.494}_{-0.437}$	$-3.117^{+0.191}_{-0.202}$	$-3.296^{+0.208}_{-0.232}$	$2.394^{+0.014}_{-0.014}$	$-2.204^{+0.005}_{-0.005}$	$1.555^{+0.008}_{-0.008}$	$0.132^{+0.013}_{-0.013}$
	QJ+WP+Pl	EE	$7.418^{+0.294}_{-0.287}$	$-3.623^{+0.042}_{-0.042}$	$-3.088^{+0.019}_{-0.020}$	$3.181^{+0.019}_{-0.019}$	$-2.542^{+0.006}_{-0.006}$	$1.521^{+0.008}_{-0.007}$	$0.095^{+0.005}_{-0.005}$
		BB	$1.703^{+0.206}_{-0.200}$	$-3.291^{+0.122}_{-0.131}$	$-2.960^{+0.061}_{-0.061}$	$2.395^{+0.014}_{-0.014}$	$-2.204^{+0.005}_{-0.005}$	$1.556^{+0.008}_{-0.008}$	$0.114^{+0.011}_{-0.011}$
$\lvert b\rvert > 15^\circ$	WP+Pl	EE	$5.727^{+0.580}_{-0.549}$	$-3.875^{+0.095}_{-0.100}$	$-3.182^{+0.073}_{-0.076}$	$2.066^{+0.017}_{-0.017}$	$-2.514^{+0.008}_{-0.008}$	$1.535^{+0.011}_{-0.011}$	$0.081^{+0.008}_{-0.008}$
		BB	$2.163^{+0.568}_{-0.498}$	$-3.180^{+0.228}_{-0.254}$	$-3.585^{+0.286}_{-0.327}$	$1.379^{+0.012}_{-0.012}$	$-2.323^{+0.008}_{-0.008}$	$1.539^{+0.012}_{-0.011}$	$0.123^{+0.015}_{-0.015}$
	QJ+WP+Pl	EE	$5.998^{+0.294}_{-0.287}$	$-3.764^{+0.051}_{-0.052}$	$-3.086^{+0.021}_{-0.022}$	$2.066^{+0.017}_{-0.017}$	$-2.514^{+0.008}_{-0.008}$	$1.534^{+0.011}_{-0.011}$	$0.068^{+0.006}_{-0.007}$
		BB	$1.363^{+0.218}_{-0.208}$	$-3.405^{+0.157}_{-0.171}$	$-3.003^{+0.072}_{-0.074}$	$1.380^{+0.012}_{-0.012}$	$-2.323^{+0.008}_{-0.008}$	$1.541^{+0.011}_{-0.011}$	$0.118^{+0.013}_{-0.013}$
$\lvert b\rvert > 20^\circ$	WP+Pl	EE	$6.864^{+0.654}_{-0.628}$	$-3.717^{+0.092}_{-0.098}$	$-3.379^{+0.091}_{-0.096}$	$1.410^{+0.015}_{-0.015}$	$-2.475^{+0.010}_{-0.010}$	$1.460^{+0.013}_{-0.013}$	$0.032^{+0.009}_{-0.009}$
		BB	$1.805^{+0.584}_{-0.492}$	$-3.421^{+0.275}_{-0.310}$	$-3.885^{+0.331}_{-0.376}$	$0.925^{+0.010}_{-0.010}$	$-2.242^{+0.010}_{-0.010}$	$1.534^{+0.016}_{-0.016}$	$0.154^{+0.019}_{-0.018}$
	QJ+WP+Pl	EE	$5.908^{+0.292}_{-0.286}$	$-3.726^{+0.052}_{-0.053}$	$-3.129^{+0.024}_{-0.024}$	$1.411^{+0.015}_{-0.015}$	$-2.475^{+0.010}_{-0.010}$	$1.461^{+0.013}_{-0.013}$	$0.022^{+0.007}_{-0.007}$
		BB	$1.358^{+0.235}_{-0.220}$	$-3.326^{+0.170}_{-0.183}$	$-2.978^{+0.075}_{-0.077}$	$0.926^{+0.011}_{-0.010}$	$-2.242^{+0.010}_{-0.010}$	$1.539^{+0.016}_{-0.016}$	$0.123^{+0.015}_{-0.015}$

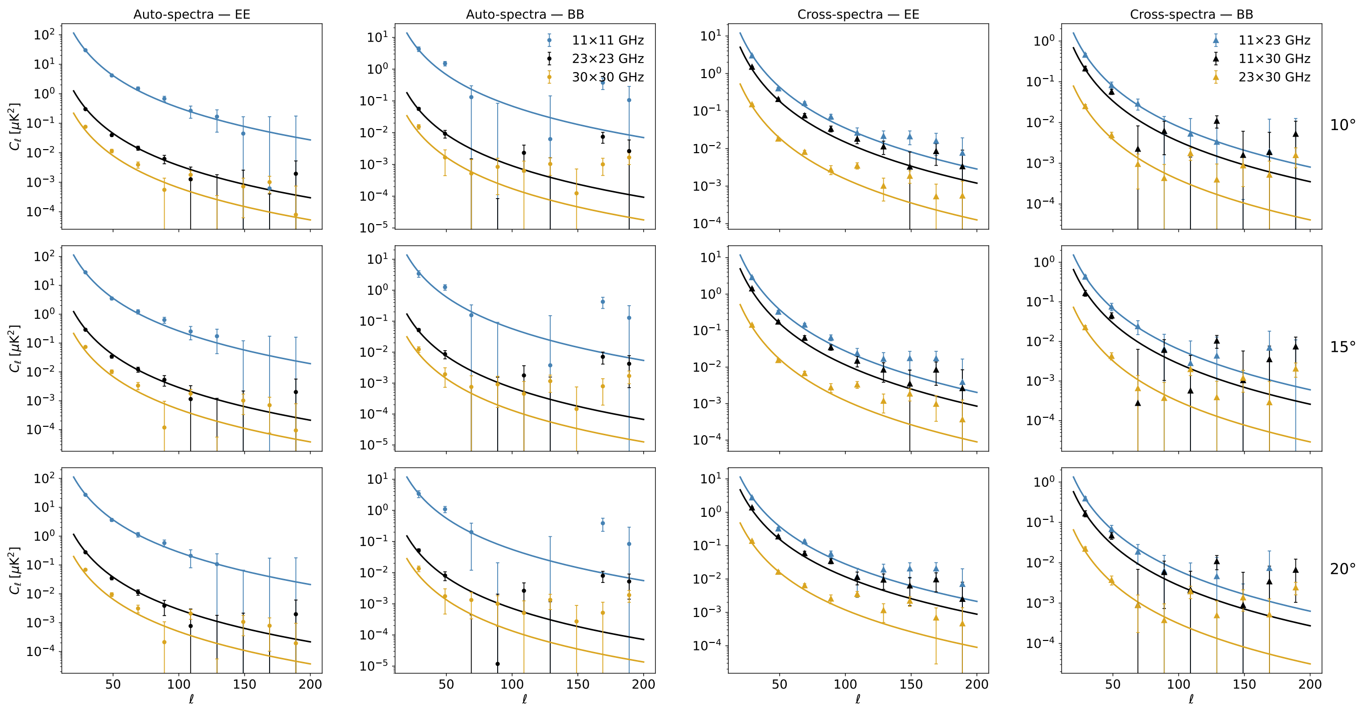

Figure 9: Measured auto- and cross-power spectra in the low-frequency bands (11, 23 and 30 GHz) for three Galactic masks. Each row corresponds to a mask (top to bottom: $|b| > 10^\circ$, 15°, 20°) and the four columns show, from left to right, auto-spectra EE, auto-spectra BB, cross-spectra EE and cross-spectra BB. Markers are binned band powers C_ℓ with 1σ uncertainties (circles for auto-spectra, and triangles for cross-spectra). Solid lines show the best-fit full model from Equation 16 using only cross data, and evaluated for each band pair.

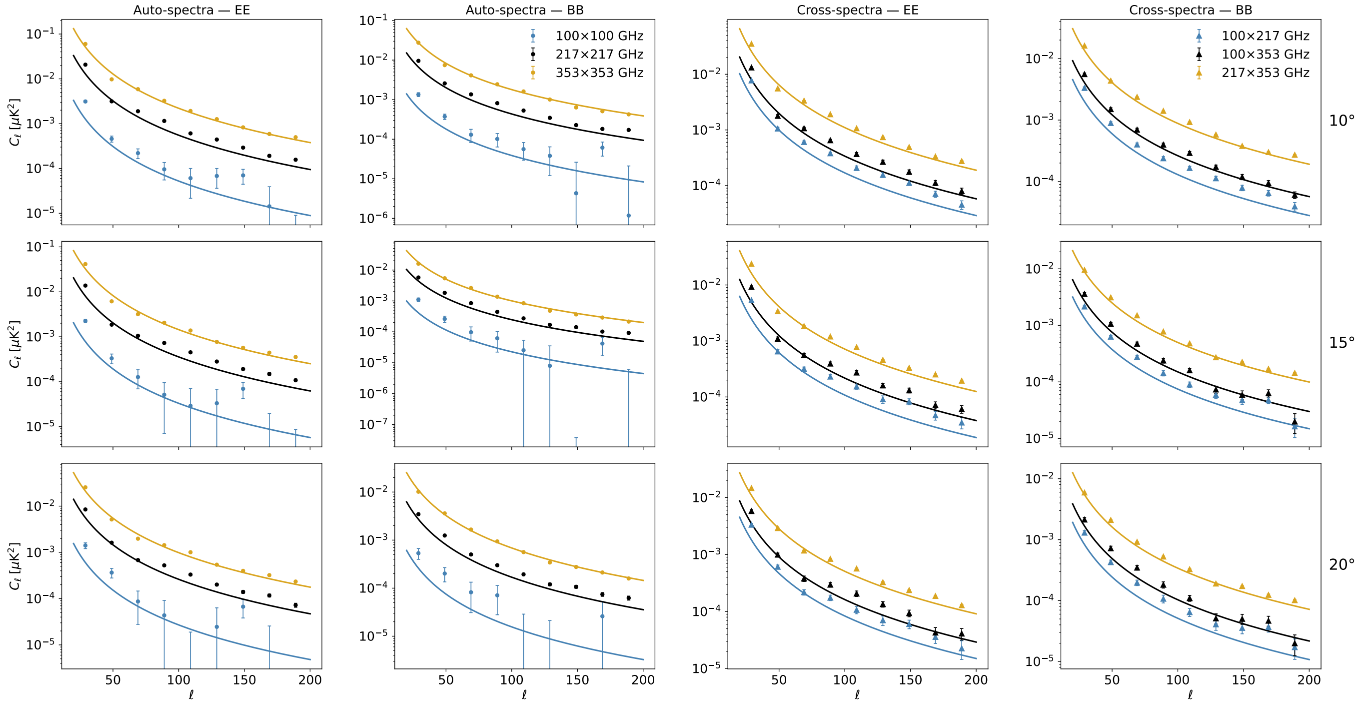

Figure 10: Measured auto- and cross-power spectra in the high-frequency bands (100, 217 and 353 GHz) for three Galactic masks. Each row corresponds to a mask (top to bottom: $|b| > 10°$, $15°$, $20°$) and the four columns show, from left to right, auto-spectra EE, auto-spectra BB, cross-spectra EE and cross-spectra BB. Markers are binned band powers C_ℓ with 1σ uncertainties (circles for auto-spectra, and triangles for cross-spectra). Solid lines show the best-fit full model from Equation 16 using only cross data, and evaluated for each band pair.

Table 6: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and the synchrotron-dust correlation coefficient ρ. The parameters are fitted independently in each multipole bin from the EE (top) and BB (bottom) polarization cross-spectra of the 14-band QJ+WP+Pl dataset, using the $|b| > 10°$ Galactic mask.

ℓ range	ℓ_{eff}	$A_{\mathrm{s}}^{\mathrm{EE}}$ [$10^{-3}\,\mu\mathrm{K}^2$]	$\beta_{\mathrm{s}}^{\mathrm{EE}}$	$A_{\mathrm{d}}^{\mathrm{EE}}$ [$10^{-3}\,\mu\mathrm{K}^2$]	$\beta_{\mathrm{d}}^{\mathrm{EE}}$	ρ^{EE}	χ^2_{red}
20–39	29.0	$296.777^{+3.753}_{-3.784}$	$-3.078^{+0.022}_{-0.022}$	$48.896^{+0.490}_{-0.485}$	$1.477^{+0.011}_{-0.011}$	$0.091^{+0.007}_{-0.007}$	1.328
40–59	49.0	$37.435^{+1.334}_{-1.339}$	$-3.148^{+0.057}_{-0.057}$	$7.813^{+0.126}_{-0.125}$	$1.507^{+0.019}_{-0.019}$	$0.117^{+0.013}_{-0.012}$	1.186
60–79	69.0	$12.839^{+0.855}_{-0.888}$	$-3.304^{+0.095}_{-0.096}$	$4.820^{+0.080}_{-0.080}$	$1.537^{+0.020}_{-0.019}$	$0.046^{+0.014}_{-0.014}$	1.072
80–99	89.0	$6.384^{+0.635}_{-0.652}$	$-3.126^{+0.133}_{-0.135}$	$2.703^{+0.041}_{-0.040}$	$1.525^{+0.021}_{-0.020}$	$0.055^{+0.020}_{-0.019}$	1.476
100–119	109.0	$3.714^{+0.531}_{-0.554}$	$-3.013^{+0.179}_{-0.185}$	$1.529^{+0.033}_{-0.033}$	$1.550^{+0.031}_{-0.030}$	$0.033^{+0.026}_{-0.026}$	0.968
120–139	129.0	$2.833^{+0.514}_{-0.534}$	$-2.582^{+0.216}_{-0.215}$	$0.977^{+0.024}_{-0.023}$	$1.368^{+0.034}_{-0.032}$	$0.156^{+0.040}_{-0.038}$	1.457
140–159	149.0	$2.106^{+0.607}_{-0.574}$	$-2.974^{+0.237}_{-0.248}$	$0.677^{+0.017}_{-0.016}$	$1.472^{+0.039}_{-0.038}$	$0.218^{+0.052}_{-0.044}$	1.037
160–179	169.0	$1.386^{+0.527}_{-0.522}$	$-3.168^{+0.298}_{-0.283}$	$0.472^{+0.017}_{-0.016}$	$1.522^{+0.050}_{-0.047}$	$0.072^{+0.065}_{-0.056}$	1.036
180–199	189.0	$1.295^{+0.689}_{-0.671}$	$-2.902^{+0.294}_{-0.306}$	$0.413^{+0.017}_{-0.016}$	$1.685^{+0.062}_{-0.062}$	$0.113^{+0.086}_{-0.063}$	1.236
ℓ range	ℓ_{eff}	$A_{\mathrm{s}}^{\mathrm{BB}}$ [$10^{-3}\,\mu\mathrm{K}^2$]	$\beta_{\mathrm{s}}^{\mathrm{BB}}$	$A_{\mathrm{d}}^{\mathrm{BB}}$ [$10^{-3}\,\mu\mathrm{K}^2$]	$\beta_{\mathrm{d}}^{\mathrm{BB}}$	ρ^{BB}	χ^2_{red}
20–39	29.0	$46.872^{+2.152}_{-2.089}$	$-3.014^{+0.069}_{-0.070}$	$22.853^{+0.315}_{-0.301}$	$1.526^{+0.017}_{-0.016}$	$0.120^{+0.014}_{-0.014}$	1.340
40–59	49.0	$9.530^{+0.800}_{-0.804}$	$-2.969^{+0.122}_{-0.123}$	$6.130^{+0.094}_{-0.092}$	$1.503^{+0.019}_{-0.019}$	$0.111^{+0.022}_{-0.021}$	1.318
60–79	69.0	$1.411^{+0.534}_{-0.535}$	$-3.154^{+0.254}_{-0.258}$	$3.440^{+0.052}_{-0.052}$	$1.573^{+0.020}_{-0.020}$	$0.083^{+0.047}_{-0.038}$	1.616
80–99	89.0	$0.935^{+0.453}_{-0.451}$	$-3.013^{+0.232}_{-0.238}$	$2.048^{+0.031}_{-0.032}$	$1.584^{+0.023}_{-0.022}$	$0.139^{+0.072}_{-0.049}$	1.429
100–119	109.0	$0.792^{+0.476}_{-0.468}$	$-2.733^{+0.251}_{-0.270}$	$1.355^{+0.024}_{-0.023}$	$1.605^{+0.025}_{-0.025}$	$0.113^{+0.084}_{-0.051}$	1.217
120–139	129.0	$1.176^{+0.427}_{-0.424}$	$-3.019^{+0.246}_{-0.253}$	$0.832^{+0.016}_{-0.015}$	$1.559^{+0.029}_{-0.028}$	$0.137^{+0.055}_{-0.043}$	1.607
140–159	149.0	$0.320^{+0.495}_{-0.268}$	$-2.952^{+0.317}_{-0.314}$	$0.541^{+0.014}_{-0.014}$	$1.555^{+0.039}_{-0.038}$	$-0.016^{+0.098}_{-0.104}$	1.156
160–179	169.0	$0.536^{+0.505}_{-0.413}$	$-3.063^{+0.308}_{-0.308}$	$0.439^{+0.013}_{-0.013}$	$1.606^{+0.045}_{-0.044}$	$-0.064^{+0.074}_{-0.103}$	1.410
180–199	189.0	$0.794^{+0.586}_{-0.506}$	$-3.283^{+0.307}_{-0.296}$	$0.404^{+0.011}_{-0.010}$	$1.651^{+0.043}_{-0.043}$	$0.103^{+0.097}_{-0.064}$	1.375

Table 7: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and the synchrotron-dust correlation coefficient ρ. The parameters are fitted independently in each multipole bin from the EE (top) and BB (bottom) polarization cross-spectra of the 14-band QJ+WP+Pl dataset, using the $|b| > 15°$ Galactic mask.

ℓ range	ℓ_{eff}	$A_{\mathrm{s}}^{\mathrm{EE}}$ [$10^{-3}\,\mu\mathrm{K}^2$]	$\beta_{\mathrm{s}}^{\mathrm{EE}}$	$A_{\mathrm{d}}^{\mathrm{EE}}$ [$10^{-3}\,\mu\mathrm{K}^2$]	$\beta_{\mathrm{d}}^{\mathrm{EE}}$	ρ^{EE}	χ^2_{red}
20–39	29.0	$276.030^{+3.804}_{-3.920}$	$-3.088^{+0.023}_{-0.023}$	$32.899^{+0.460}_{-0.452}$	$1.443^{+0.016}_{-0.016}$	$0.077^{+0.008}_{-0.008}$	1.573
40–59	49.0	$31.665^{+1.343}_{-1.301}$	$-3.115^{+0.068}_{-0.069}$	$4.864^{+0.121}_{-0.118}$	$1.565^{+0.030}_{-0.030}$	$0.046^{+0.015}_{-0.015}$	0.975
60–79	69.0	$10.732^{+0.864}_{-0.867}$	$-3.327^{+0.111}_{-0.115}$	$2.644^{+0.060}_{-0.059}$	$1.529^{+0.030}_{-0.028}$	$-0.004^{+0.018}_{-0.018}$	1.077
80–99	89.0	$6.012^{+0.646}_{-0.648}$	$-3.153^{+0.137}_{-0.143}$	$1.708^{+0.036}_{-0.035}$	$1.541^{+0.029}_{-0.028}$	$0.079^{+0.021}_{-0.021}$	1.461
100–119	109.0	$3.355^{+0.532}_{-0.520}$	$-2.925^{+0.179}_{-0.186}$	$1.096^{+0.029}_{-0.028}$	$1.530^{+0.037}_{-0.037}$	$0.093^{+0.029}_{-0.028}$	1.239
120–139	129.0	$2.466^{+0.557}_{-0.563}$	$-2.738^{+0.246}_{-0.250}$	$0.600^{+0.019}_{-0.019}$	$1.333^{+0.043}_{-0.041}$	$0.109^{+0.045}_{-0.042}$	1.401
140–159	149.0	$2.033^{+0.648}_{-0.631}$	$-2.921^{+0.275}_{-0.280}$	$0.450^{+0.014}_{-0.014}$	$1.443^{+0.051}_{-0.048}$	$0.142^{+0.054}_{-0.049}$	1.114
160–179	169.0	$1.723^{+0.578}_{-0.562}$	$-3.042^{+0.271}_{-0.282}$	$0.369^{+0.016}_{-0.015}$	$1.598^{+0.064}_{-0.061}$	$0.035^{+0.058}_{-0.055}$	0.983
180–199	189.0	$0.969^{+0.733}_{-0.658}$	$-2.901^{+0.307}_{-0.307}$	$0.295^{+0.016}_{-0.015}$	$1.716^{+0.090}_{-0.086}$	$0.103^{+0.119}_{-0.078}$	1.239
ℓ range	ℓ_{eff}	$A_{\mathrm{s}}^{\mathrm{BB}}$ [$10^{-3}\,\mu\mathrm{K}^2$]	$\beta_{\mathrm{s}}^{\mathrm{BB}}$	$A_{\mathrm{d}}^{\mathrm{BB}}$ [$10^{-3}\,\mu\mathrm{K}^2$]	$\beta_{\mathrm{d}}^{\mathrm{BB}}$	ρ^{BB}	χ^2_{red}
20–39	29.0	$40.394^{+2.272}_{-2.243}$	$-3.130^{+0.083}_{-0.085}$	$13.261^{+0.253}_{-0.252}$	$1.497^{+0.022}_{-0.022}$	$0.168^{+0.017}_{-0.017}$	1.284
40–59	49.0	$8.505^{+0.860}_{-0.893}$	$-2.987^{+0.152}_{-0.146}$	$4.382^{+0.082}_{-0.086}$	$1.492^{+0.024}_{-0.023}$	$0.045^{+0.024}_{-0.023}$	1.596
60–79	69.0	$1.059^{+0.551}_{-0.556}$	$-3.099^{+0.261}_{-0.263}$	$2.176^{+0.043}_{-0.043}$	$1.586^{+0.027}_{-0.027}$	$0.112^{+0.078}_{-0.053}$	1.476
80–99	89.0	$0.933^{+0.494}_{-0.506}$	$-2.873^{+0.246}_{-0.262}$	$1.114^{+0.027}_{-0.027}$	$1.557^{+0.035}_{-0.035}$	$0.113^{+0.086}_{-0.058}$	1.275
100–119	109.0	$1.052^{+0.498}_{-0.501}$	$-2.853^{+0.278}_{-0.288}$	$0.694^{+0.018}_{-0.018}$	$1.592^{+0.040}_{-0.039}$	$0.038^{+0.056}_{-0.049}$	1.047
120–139	129.0	$1.146^{+0.443}_{-0.439}$	$-3.103^{+0.279}_{-0.281}$	$0.391^{+0.015}_{-0.014}$	$1.567^{+0.058}_{-0.056}$	$0.082^{+0.062}_{-0.050}$	1.331
140–159	149.0	$0.377^{+0.532}_{-0.310}$	$-2.953^{+0.307}_{-0.305}$	$0.335^{+0.014}_{-0.013}$	$1.660^{+0.067}_{-0.064}$	$0.032^{+0.119}_{-0.092}$	1.116
160–179	169.0	$0.610^{+0.509}_{-0.437}$	$-3.186^{+0.292}_{-0.278}$	$0.238^{+0.012}_{-0.011}$	$1.527^{+0.076}_{-0.072}$	$-0.126^{+0.085}_{-0.137}$	1.296
180–199	189.0	$1.485^{+0.677}_{-0.689}$	$-2.987^{+0.263}_{-0.280}$	$0.225^{+0.012}_{-0.011}$	$1.786^{+0.087}_{-0.082}$	$0.029^{+0.066}_{-0.061}$	1.123

Table 8: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and the synchrotron-dust correlation coefficient ρ. The parameters are fitted independently in each multipole bin from the EE (top) and BB (bottom) polarization cross-spectra of the 14-band QJ+WP+Pl dataset, using the $|b| > 20^\circ$ Galactic mask.

ℓ range	$\ell_{\rm eff}$	$A_{\rm s}^{\rm EE}\,[10^{-3}\,\mu{\rm K}^2]$	$\beta_{\rm s}^{\rm EE}$	$A_{\rm d}^{\rm EE}\,[10^{-3}\,\mu{\rm K}^2]$	$\beta_{\rm d}^{\rm EE}$	$\rho^{\rm EE}$	$\chi^2_{\rm red}$
20–39	29.0	$264.245^{+4.366}_{-4.499}$	$-3.116^{+0.027}_{-0.027}$	$20.371^{+0.406}_{-0.402}$	$1.456^{+0.021}_{-0.022}$	$0.009^{+0.010}_{-0.010}$	1.518
40–59	49.0	$31.624^{+1.382}_{-1.419}$	$-3.125^{+0.069}_{-0.071}$	$4.020^{+0.105}_{-0.106}$	$1.472^{+0.031}_{-0.031}$	$0.066^{+0.015}_{-0.015}$	1.503
60–79	69.0	$10.313^{+0.794}_{-0.813}$	$-3.266^{+0.112}_{-0.120}$	$1.631^{+0.050}_{-0.049}$	$1.446^{+0.039}_{-0.039}$	$-0.088^{+0.021}_{-0.022}$	0.992
80–99	89.0	$5.609^{+0.661}_{-0.676}$	$-3.129^{+0.153}_{-0.156}$	$1.172^{+0.034}_{-0.033}$	$1.493^{+0.040}_{-0.038}$	$0.062^{+0.023}_{-0.024}$	1.331
100–119	109.0	$2.531^{+0.546}_{-0.545}$	$-2.962^{+0.215}_{-0.224}$	$0.812^{+0.029}_{-0.029}$	$1.556^{+0.053}_{-0.050}$	$0.039^{+0.037}_{-0.035}$	1.110
120–139	129.0	$2.210^{+0.555}_{-0.543}$	$-2.773^{+0.250}_{-0.247}$	$0.412^{+0.017}_{-0.016}$	$1.240^{+0.057}_{-0.053}$	$0.163^{+0.056}_{-0.049}$	1.305
140–159	149.0	$2.518^{+0.630}_{-0.617}$	$-2.777^{+0.236}_{-0.252}$	$0.321^{+0.014}_{-0.014}$	$1.432^{+0.067}_{-0.065}$	$0.094^{+0.050}_{-0.047}$	1.120
160–179	169.0	$1.884^{+0.598}_{-0.566}$	$-3.110^{+0.293}_{-0.289}$	$0.274^{+0.015}_{-0.014}$	$1.611^{+0.084}_{-0.080}$	$0.038^{+0.059}_{-0.058}$	1.063
180–199	189.0	$0.973^{+0.714}_{-0.648}$	$-2.899^{+0.295}_{-0.303}$	$0.191^{+0.015}_{-0.013}$	$1.666^{+0.136}_{-0.125}$	$0.117^{+0.130}_{-0.094}$	1.099
ℓ range	$\ell_{\rm eff}$	$A_{\rm s}^{\rm BB}\,[10^{-3}\,\mu{\rm K}^2]$	$\beta_{\rm s}^{\rm BB}$	$A_{\rm d}^{\rm BB}\,[10^{-3}\,\mu{\rm K}^2]$	$\beta_{\rm d}^{\rm BB}$	$\rho^{\rm BB}$	$\chi^2_{\rm red}$
20–39	29.0	$37.722^{+2.289}_{-2.316}$	$-3.070^{+0.085}_{-0.088}$	$8.376^{+0.211}_{-0.209}$	$1.523^{+0.032}_{-0.032}$	$0.177^{+0.021}_{-0.020}$	1.741
40–59	49.0	$7.944^{+0.884}_{-0.899}$	$-3.016^{+0.162}_{-0.160}$	$2.934^{+0.071}_{-0.068}$	$1.480^{+0.031}_{-0.029}$	$0.036^{+0.027}_{-0.027}$	1.258
60–79	69.0	$0.949^{+0.602}_{-0.586}$	$-2.977^{+0.278}_{-0.275}$	$1.318^{+0.038}_{-0.037}$	$1.568^{+0.042}_{-0.040}$	$0.174^{+0.129}_{-0.072}$	1.527
80–99	89.0	$0.906^{+0.528}_{-0.534}$	$-2.941^{+0.262}_{-0.271}$	$0.755^{+0.026}_{-0.025}$	$1.568^{+0.051}_{-0.048}$	$0.066^{+0.083}_{-0.065}$	1.670
100–119	109.0	$1.235^{+0.538}_{-0.528}$	$-2.874^{+0.283}_{-0.295}$	$0.466^{+0.016}_{-0.016}$	$1.543^{+0.052}_{-0.051}$	$0.049^{+0.060}_{-0.054}$	1.043
120–139	129.0	$1.116^{+0.468}_{-0.484}$	$-3.117^{+0.285}_{-0.279}$	$0.278^{+0.014}_{-0.014}$	$1.614^{+0.084}_{-0.082}$	$0.097^{+0.076}_{-0.060}$	1.160
140–159	149.0	$0.457^{+0.599}_{-0.375}$	$-2.938^{+0.305}_{-0.304}$	$0.272^{+0.014}_{-0.014}$	$1.794^{+0.091}_{-0.087}$	$0.017^{+0.115}_{-0.098}$	1.128
160–179	169.0	$0.840^{+0.578}_{-0.542}$	$-3.096^{+0.274}_{-0.276}$	$0.173^{+0.011}_{-0.010}$	$1.523^{+0.102}_{-0.095}$	$-0.103^{+0.080}_{-0.111}$	1.111
180–199	189.0	$1.683^{+0.697}_{-0.731}$	$-2.855^{+0.245}_{-0.271}$	$0.154^{+0.011}_{-0.010}$	$1.706^{+0.123}_{-0.113}$	$-0.054^{+0.069}_{-0.073}$	1.158

B Alternative multipole fitting range $30 \leq \ell \leq 200$

This appendix presents the results of the global power-law fitting analysis carried out over the multipole range $30 \leq \ell \leq 200$, as a robustness check against the baseline range $20 \leq \ell \leq 200$ adopted in Section 5. Starting at $\ell = 30$ is the conventional choice in the literature (Rubiño-Martín et al., 2023; Martire et al., 2022), as it avoids the largest angular scales where residual systematic effects and foreground modelling uncertainties may be more significant.

However, as discussed in Section 5, excluding the first multipole bin ($20 \leq \ell \leq 39$, $\ell_{\rm eff} = 29$) has a marked impact on the BB-mode power-law fit. The BB polarization power spectrum is intrinsically fainter than EE and falls steeply with multipole, as a result, the first bin carries a substantial fraction of the total constraining power on the BB angular spectral index α_s^{BB}. Without it, the remaining bins span a narrower dynamic range in signal amplitude, and the power-law slope becomes poorly constrained. This leads to an artificial steepening of the inferred α_s^{BB} towards unphysically large negative values, accompanied by significantly enlarged posterior uncertainties. The effect is most pronounced in the $|b| > 10°$ mask when QUIJOTE data are included, where the additional low-frequency sensitivity of QUIJOTE makes the model sensitive to the shape of the synchrotron angular spectrum, making the results more dependent on the missing low-ℓ information.

The results below are presented for both the auto+cross-spectra and cross-spectra-only configurations, and for the three Galactic masks, to allow a direct comparison with the baseline results of Section 5 and Appendix A.

Comparing Table 9 with the baseline Table 1, the most prominent change when the first bin is removed concerns the synchrotron angular spectral index α_s. For the EE mode, the QJ+WP+Pl estimates flatten from $\alpha_s^{EE} \approx -3.6$ to -3.7 in the baseline to -2.9 to -3.0 here, in better agreement with literature values (Martire et al., 2022; Krachmalnicoff et al., 2018). For the BB mode, however, the effect is the opposite, as the inferred slope steepens to $\alpha_s^{BB} \approx -3.6$ to -4.1 (QJ+WP+Pl), with uncertainties two to three times larger than in the baseline. This is the direct consequence of the loss of the first multipole bin, which carries a disproportionate fraction of the total BB constraining power, as discussed in Section 5.1.2. The remaining parameters are stable between the two multipole ranges and are mutually consistent within 1-2σ, confirming that the sensitivity to the fitting range is largely confined to the angular spectral index.

The cross-spectra-only results at $30 \leq \ell \leq 200$ (Table 10) show the same qualitative pattern as the corresponding auto+cross results. The EE slopes flatten relative to the cross-only baseline (Table 5, Appendix A), while the BB slopes become steeper and significantly less well constrained. Across both configurations, the WP+Pl-only BB fits at $30 \leq \ell \leq 200$ are particularly unreliable, with slope uncertainties of ~ 0.5-0.9. Adding QUIJOTE data reduces the BB errors substantially, but the posteriors remain broader than in the baseline. The dust parameters and β_s are again stable, reinforcing the conclusion that the choice of lower multipole limit primarily affects the synchrotron angular spectral index.

Table 9: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and synchrotron-dust correlation coefficient ρ, from a global power-law fit to the EE and BB polarization auto- and cross-spectra of the WP+Pl and QJ+WP+Pl datasets over the multipole range $30 \leq \ell \leq 200$, for the three Galactic masks.

Mask	Dataset	Mode	$A_s\,[10^{-3}\,\mu\mathrm{K}^2]$	α_s	β_s	$A_d\,[10^{-3}\,\mu\mathrm{K}^2]$	α_d	β_d	ρ
$\lvert b\rvert > 10°$	WP+Pl	EE	$9.451^{+0.649}_{-0.644}$	$-2.845^{+0.146}_{-0.156}$	$-3.189^{+0.116}_{-0.120}$	$2.999^{+0.012}_{-0.012}$	$-2.244^{+0.006}_{-0.006}$	$1.523^{+0.006}_{-0.006}$	$0.090^{+0.011}_{-0.011}$
		BB	$2.589^{+0.534}_{-0.530}$	$-2.380^{+0.443}_{-0.506}$	$-3.402^{+0.295}_{-0.325}$	$2.336^{+0.008}_{-0.008}$	$-2.163^{+0.006}_{-0.006}$	$1.536^{+0.005}_{-0.005}$	$0.125^{+0.020}_{-0.019}$
	QJ+WP+Pl	EE	$8.806^{+0.325}_{-0.328}$	$-2.949^{+0.079}_{-0.083}$	$-3.148^{+0.036}_{-0.036}$	$2.999^{+0.012}_{-0.012}$	$-2.244^{+0.006}_{-0.006}$	$1.523^{+0.006}_{-0.006}$	$0.088^{+0.008}_{-0.008}$
		BB	$1.266^{+0.296}_{-0.284}$	$-4.074^{+0.449}_{-0.535}$	$-3.065^{+0.109}_{-0.110}$	$2.336^{+0.008}_{-0.008}$	$-2.163^{+0.006}_{-0.006}$	$1.535^{+0.005}_{-0.005}$	$0.103^{+0.016}_{-0.015}$
$\lvert b\rvert > 15°$	WP+Pl	EE	$8.697^{+0.678}_{-0.677}$	$-2.695^{+0.169}_{-0.181}$	$-3.234^{+0.138}_{-0.143}$	$1.889^{+0.011}_{-0.011}$	$-2.112^{+0.009}_{-0.009}$	$1.533^{+0.008}_{-0.008}$	$0.066^{+0.012}_{-0.012}$
		BB	$2.667^{+0.557}_{-0.569}$	$-1.946^{+0.497}_{-0.595}$	$-3.232^{+0.338}_{-0.383}$	$1.417^{+0.007}_{-0.007}$	$-2.419^{+0.009}_{-0.009}$	$1.543^{+0.008}_{-0.008}$	$0.088^{+0.021}_{-0.020}$
	QJ+WP+Pl	EE	$7.889^{+0.332}_{-0.335}$	$-2.816^{+0.094}_{-0.095}$	$-3.130^{+0.042}_{-0.042}$	$1.889^{+0.011}_{-0.011}$	$-2.111^{+0.009}_{-0.009}$	$1.533^{+0.008}_{-0.008}$	$0.051^{+0.009}_{-0.009}$
		BB	$1.266^{+0.371}_{-0.355}$	$-3.861^{+0.570}_{-0.697}$	$-2.986^{+0.136}_{-0.135}$	$1.416^{+0.007}_{-0.007}$	$-2.418^{+0.009}_{-0.009}$	$1.542^{+0.008}_{-0.008}$	$0.062^{+0.018}_{-0.017}$
$\lvert b\rvert > 20°$	WP+Pl	EE	$7.902^{+0.720}_{-0.716}$	$-2.957^{+0.203}_{-0.216}$	$-3.260^{+0.145}_{-0.151}$	$1.362^{+0.010}_{-0.010}$	$-2.205^{+0.011}_{-0.011}$	$1.487^{+0.011}_{-0.011}$	$0.056^{+0.013}_{-0.013}$
		BB	$2.675^{+0.588}_{-0.613}$	$-1.663^{+0.543}_{-0.661}$	$-3.055^{+0.350}_{-0.403}$	$0.949^{+0.006}_{-0.006}$	$-2.335^{+0.011}_{-0.011}$	$1.540^{+0.011}_{-0.010}$	$0.079^{+0.025}_{-0.023}$
	QJ+WP+Pl	EE	$7.218^{+0.349}_{-0.347}$	$-3.010^{+0.108}_{-0.111}$	$-3.134^{+0.044}_{-0.044}$	$1.362^{+0.010}_{-0.010}$	$-2.205^{+0.011}_{-0.011}$	$1.486^{+0.011}_{-0.011}$	$0.033^{+0.010}_{-0.010}$
		BB	$1.369^{+0.408}_{-0.401}$	$-3.561^{+0.601}_{-0.744}$	$-2.918^{+0.152}_{-0.152}$	$0.949^{+0.006}_{-0.006}$	$-2.335^{+0.011}_{-0.011}$	$1.539^{+0.010}_{-0.010}$	$0.066^{+0.021}_{-0.020}$

Table 10: Posterior constraints (median and 68% credible intervals) on the synchrotron and dust amplitudes, spectral indices, and synchrotron-dust correlation coefficient ρ, from a global power-law fit to the EE and BB polarization cross-spectra of the WP+Pl and QJ+WP+Pl datasets over the multipole range $30 \leq \ell \leq 200$, for the three Galactic masks.

Mask	Dataset	Mode	$A_s\,[10^{-3}\,\mu\mathrm{K}^2]$	α_s	β_s	$A_d\,[10^{-3}\,\mu\mathrm{K}^2]$	α_d	β_d	ρ
$\lvert b\rvert > 10°$	WP+Pl	EE	$11.794^{+1.193}_{-1.126}$	$-2.661^{+0.164}_{-0.172}$	$-3.550^{+0.203}_{-0.225}$	$2.962^{+0.022}_{-0.022}$	$-2.240^{+0.008}_{-0.008}$	$1.512^{+0.010}_{-0.009}$	$0.088^{+0.010}_{-0.010}$
		BB	$3.854^{+1.018}_{-0.885}$	$-2.075^{+0.402}_{-0.447}$	$-3.981^{+0.466}_{-0.553}$	$2.372^{+0.016}_{-0.016}$	$-2.160^{+0.008}_{-0.008}$	$1.561^{+0.009}_{-0.009}$	$0.113^{+0.019}_{-0.017}$
	QJ+WP+Pl	EE	$9.027^{+0.360}_{-0.362}$	$-2.868^{+0.089}_{-0.090}$	$-3.170^{+0.046}_{-0.047}$	$2.967^{+0.022}_{-0.022}$	$-2.240^{+0.008}_{-0.008}$	$1.515^{+0.010}_{-0.009}$	$0.087^{+0.008}_{-0.008}$
		BB	$1.642^{+0.319}_{-0.321}$	$-3.482^{+0.406}_{-0.470}$	$-2.891^{+0.120}_{-0.119}$	$2.373^{+0.016}_{-0.016}$	$-2.159^{+0.008}_{-0.008}$	$1.564^{+0.009}_{-0.009}$	$0.106^{+0.016}_{-0.015}$
$\lvert b\rvert > 15°$	WP+Pl	EE	$11.700^{+1.432}_{-1.278}$	$-2.396^{+0.177}_{-0.190}$	$-3.737^{+0.265}_{-0.306}$	$1.872^{+0.019}_{-0.019}$	$-2.110^{+0.011}_{-0.011}$	$1.525^{+0.014}_{-0.013}$	$0.066^{+0.011}_{-0.011}$
		BB	$4.234^{+1.211}_{-1.041}$	$-1.638^{+0.464}_{-0.515}$	$-4.233^{+0.637}_{-0.770}$	$1.422^{+0.014}_{-0.013}$	$-2.420^{+0.012}_{-0.012}$	$1.551^{+0.013}_{-0.013}$	$0.081^{+0.019}_{-0.017}$
	QJ+WP+Pl	EE	$8.104^{+0.364}_{-0.363}$	$-2.706^{+0.100}_{-0.104}$	$-3.151^{+0.055}_{-0.055}$	$1.873^{+0.019}_{-0.019}$	$-2.110^{+0.011}_{-0.011}$	$1.526^{+0.013}_{-0.013}$	$0.051^{+0.009}_{-0.009}$
		BB	$1.685^{+0.367}_{-0.387}$	$-3.171^{+0.480}_{-0.594}$	$-2.857^{+0.138}_{-0.145}$	$1.423^{+0.014}_{-0.014}$	$-2.420^{+0.012}_{-0.012}$	$1.553^{+0.013}_{-0.013}$	$0.063^{+0.017}_{-0.017}$
$\lvert b\rvert > 20°$	WP+Pl	EE	$11.296^{+1.558}_{-1.375}$	$-2.647^{+0.211}_{-0.228}$	$-3.901^{+0.304}_{-0.352}$	$1.334^{+0.017}_{-0.017}$	$-2.211^{+0.015}_{-0.014}$	$1.460^{+0.017}_{-0.017}$	$0.055^{+0.012}_{-0.012}$
		BB	$4.148^{+1.380}_{-1.156}$	$-1.387^{+0.532}_{-0.621}$	$-4.149^{+0.743}_{-0.914}$	$0.950^{+0.012}_{-0.012}$	$-2.337^{+0.015}_{-0.015}$	$1.543^{+0.018}_{-0.018}$	$0.075^{+0.021}_{-0.019}$
	QJ+WP+Pl	EE	$7.481^{+0.376}_{-0.378}$	$-2.880^{+0.115}_{-0.120}$	$-3.154^{+0.057}_{-0.057}$	$1.334^{+0.017}_{-0.017}$	$-2.210^{+0.014}_{-0.014}$	$1.461^{+0.017}_{-0.017}$	$0.032^{+0.010}_{-0.010}$
		BB	$1.760^{+0.384}_{-0.407}$	$-2.906^{+0.506}_{-0.613}$	$-2.827^{+0.154}_{-0.161}$	$0.952^{+0.012}_{-0.012}$	$-2.337^{+0.016}_{-0.016}$	$1.547^{+0.018}_{-0.018}$	$0.065^{+0.020}_{-0.020}$

C Reduced χ^2 values of the power-law fittings

Table 11: Reduced chi-squared values χ^2_{red} for all global power-law fits, combining the two multipole ranges ($20 \leq \ell \leq 200$ and $30 \leq \ell \leq 200$), both spectral datasets (auto- and cross-spectra, and cross-spectra only), both dataset combinations (WP+Pl and QJ+WP+Pl), and the three Galactic masks.

		$20 \leq \ell \leq 200$				$30 \leq \ell \leq 200$			
		Auto+Cross		Cross		Auto+Cross		Cross	
Mask	Dataset	EE	BB	EE	BB	EE	BB	EE	BB
$\lvert b\rvert > 10°$	WP+Pl	10.15	2.91	7.48	2.28	3.99	2.93	3.14	2.27
	QJ+WP+Pl	7.90	2.50	5.83	2.01	3.27	2.51	2.61	2.00
$\lvert b\rvert > 15°$	WP+Pl	8.72	1.95	6.46	1.63	2.22	1.63	1.92	1.48
	QJ+WP+Pl	6.85	1.84	5.10	1.59	1.95	1.57	1.73	1.45
$\lvert b\rvert > 20°$	WP+Pl	3.36	1.70	2.64	1.56	1.83	1.47	1.57	1.39
	QJ+WP+Pl	2.85	1.60	2.32	1.49	1.69	1.42	1.52	1.36